MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

IOC CHANNEL CONTROL MODULE, KEY CODE AWH

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Input/Output Controller (IOC) channel control module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.152 failures per million operating hours for class I or 2.560 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AWH.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I Module electrical requirements

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0.0	1.0	0.0	1.0	0 0	1 0	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests ---------						
D1-D3	Ibd	V	S	0.0	0.1	0.0	0.1	0 0	0.1	nA
D4	Ibd	V	S	0.0	1.0	0.0	1.0	0 0	1 0	nA
D5	Ibd	V	S	0.0	5.0	0.0	5.0	0.0	5.0	nA
D6	Ibd	V	S	0.0	0.1	0 0	0.1	0 0	0 1	nA
D7	Ibd	V	S	0.0	0.2	0.0	0.2	0 0	0 2	nA
D8	Ibd	V	S	0 0	0 1	0 0	0 1	0 0	0 1	nA
D9	Ibd	V	S	0.0	0.6	0.0	0.6	0 0	0.6	nA
D10-D14	Ibd	V	S	0.0	0.2	0.0	0.2	0 0	0 2	nA
D15-D19	Ibd	V	S	0.0	0.1	0.0	0.1	0 0	0 1	nA
D20-D35	Ibd	V	S	-0 02	0 12	-0 02	0.12	-0 02	0 12	nA
D36	Ibd	V	S	0.0	0.2	0.0	0.2	0 0	0 2	nA
E1-E7	Ios	VI	S	-325.0	-100 0	-325.0	-100.0	-325.0	-100. 0	nA
E8,E9	Ios	VI	S	-150.0	-60 0	-150.0	-60.0	-150.0	-60. 0	nA
E10	Ios	VI	S	-100.0	-15 0	-100.0	-15.0	-100.0	-15. 0	nA
E11	Ios	VI	S	-150.0	-60 0	-150.0	-60.0	-150.0	-60. 0	nA
E12	Ios	VI	S	-325.0	-100 0	-325.0	-100.0	-325.0	-100. 0	nA
E13	Ios	VI	S	-150.0	-60 0	-150.0	-60.0	-150.0	-60 0	nA
E14-E17	Ios	VI	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15. 0	nA
E18	Ios	VI	S	-150.0	-60 0	-150.0	-60.0	-150.0	-60. 0	nA
E19	Ios	VI	S	-260.0	-30 0	-260.0	-30.0	-260.0	-30. 0	nA
E20-E25	Ios	VI	S	-100.0	-20.0	-100.0	-20.0	-100.0	-20. 0	nA
E26	Ios	VI	S	-100.0	-40 0	-100.0	-40.0	-100 0	-40. 0	nA
E27-E50	Ios	VI	S	-150.0	-60 0	-150.0	-60.0	-150.0	-60. 0	nA
E51-E54	Ios	VI	S	-100.0	-20 0	-100.0	-20.0	-100.0	-20. 0	nA

- See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F1	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F2,F3	Iih	VII	X	-0.6	-0.02	-0.6	-0.02	-0.6	-0.02	mA
F4	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F5	Iih	VII	X	-10.0	0.25	-10.0	0.25	-10.0	0.25	mA
F6	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F7	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F8	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F9	Iih	VII	X	-9.4	0.12	-9.4	0.12	-9.4	0.12	mA
F10	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F11	Iih	VII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
F12-F14	Iih	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F15	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F16	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F17	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F18	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F19	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F20-F35	Iih	VII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
F36-F39	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F40	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F41-F44	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G1	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G2,G3	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G4	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G5	Iil	VIII	X	-10.0	0.25	-10.0	0.25	-10.0	0.25	mA
G6	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G7	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G8	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G9	Iil	VIII	X	-9.4	0.12	-9.4	0.12	-9.4	0.12	mA
G10	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G11	Iil	VIII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
G12-G14	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G15	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G16	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G17	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G18	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G19	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G20-G35	Iil	VIII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
G36-G39	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G40	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G41-G44	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
H1-H54	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J54	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L2-L10	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L11-L14	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L15	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L16	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L17	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L18	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L19,L20	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XII	S	8.4	15.6	8.4	15.6	8.4	15.6	uF
M2	PSFC	XII	S	7.0	13.0	7.0	13.0	7.0	13.0	uF
N	Icc	XIII	X	290	700	290	700	290	700	mA
P1,P2	Functional	XIV	X	See Table XIV						
Q1,Q2	Dynamic	XV	S	See Table XV						
Q3,Q4	Dynamic	XV	0	See Table XV						

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R1	C1/o	XVII	Q	2	75	2	75	2	75	pF
R2	C1/o	XVII	Q	2	125	2	125	2	125	pF
R3	C1/o	XVII	Q	2	75	2	75	2	75	pF
R4	C1/o	XVII	Q	2	50	2	50	2	50	pF
R5	C1/o	XVII	Q	10	100	10	100	10	100	pF
R6	C1/o	XVII	Q	2	75	2	75	2	75	pF
R7	C1/o	XVII	Q	4	75	4	75	4	75	pF
R8	C1/o	XVII	Q	2	75	2	75	2	75	pF
R9	C1/o	XVII	Q	12	150	12	150	12	150	pF
R10	C1/o	XVII	Q	2	75	2	75	2	75	pF
R11-R14	C1/o	XVII	Q	4	75	4	75	4	75	pF
R15	C1/o	XVII	Q	2	75	2	75	2	75	pF
R16	C1/o	XVII	Q	2	50	2	50	2	50	pF
R17	C1/o	XVII	Q	2	75	2	75	2	75	pF
R18	C1/o	XVII	Q	2	50	2	50	2	50	pF
R19	C1/o	XVII	Q	2	75	2	75	2	75	pF
R20	C1/o	XVII	Q	4	50	4	50	4	50	pF
R21	C1/o	XVII	Q	4	75	4	75	4	75	pF
R22	C1/o	XVII	Q	4	50	4	50	4	50	pF
R23-R25	C1/o	XVII	Q	4	75	4	75	4	75	pF
R26-R32	C1/o	XVII	Q	4	50	4	50	4	50	pF
R33-R44	C1/o	XVII	Q	4	75	4	75	4	75	pF
T1,T2	PSTA	XVIII	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/506-1 for class I and M28787/506-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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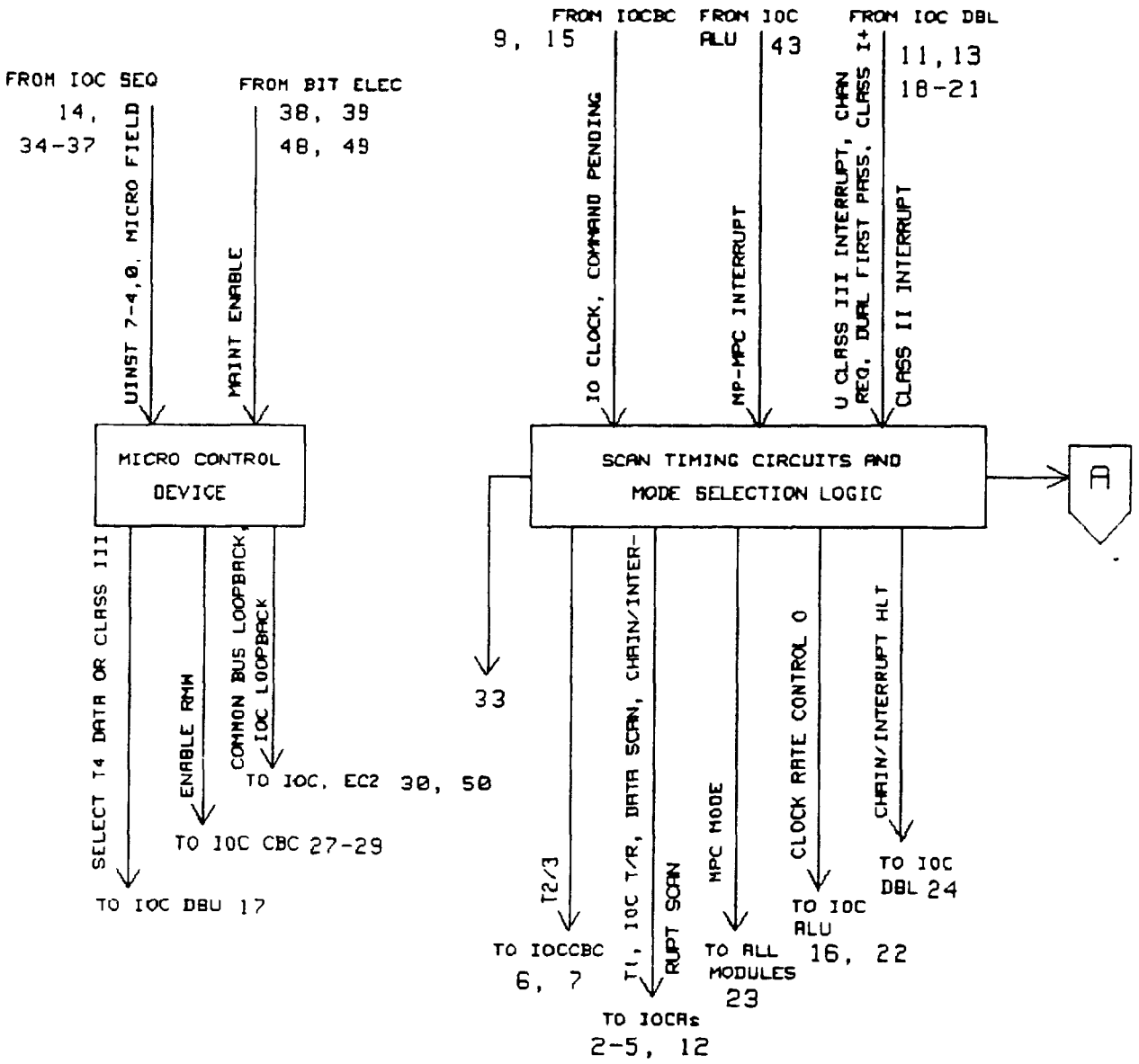


FIGURE 1. AWH functional block diagram.

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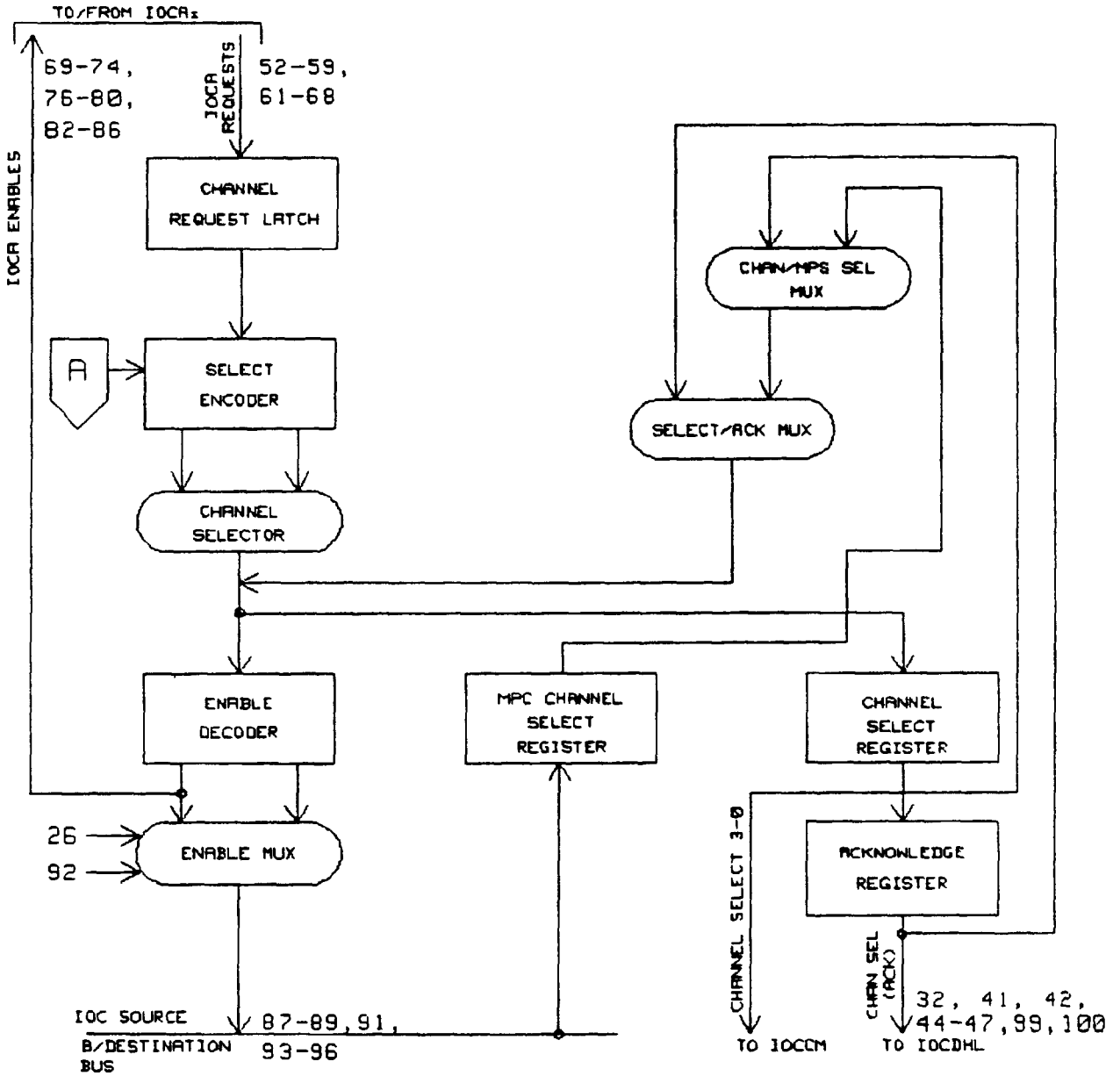


FIGURE 1. AWH funtional block diagram - Continued.

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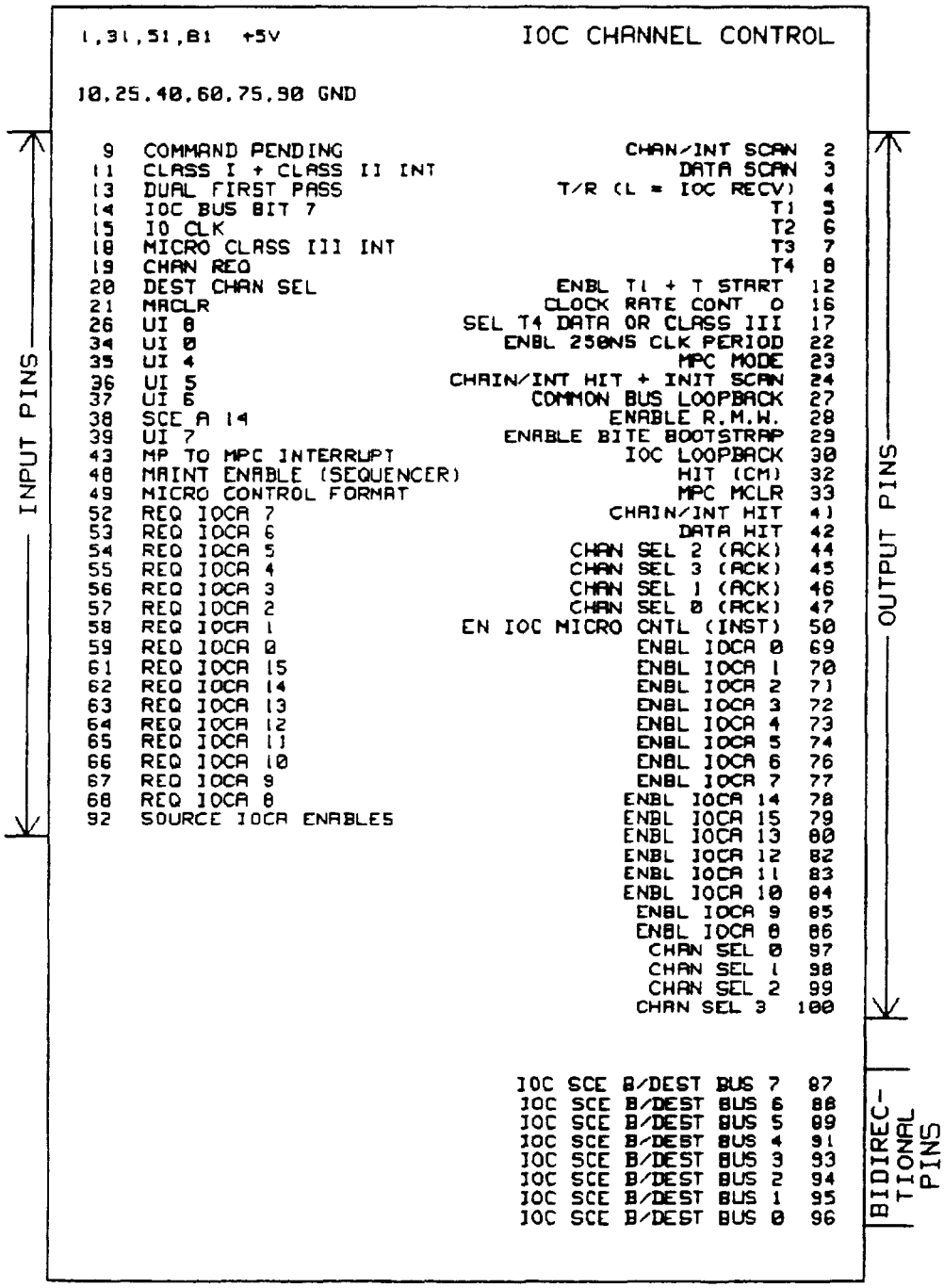


FIGURE 2. AWH pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

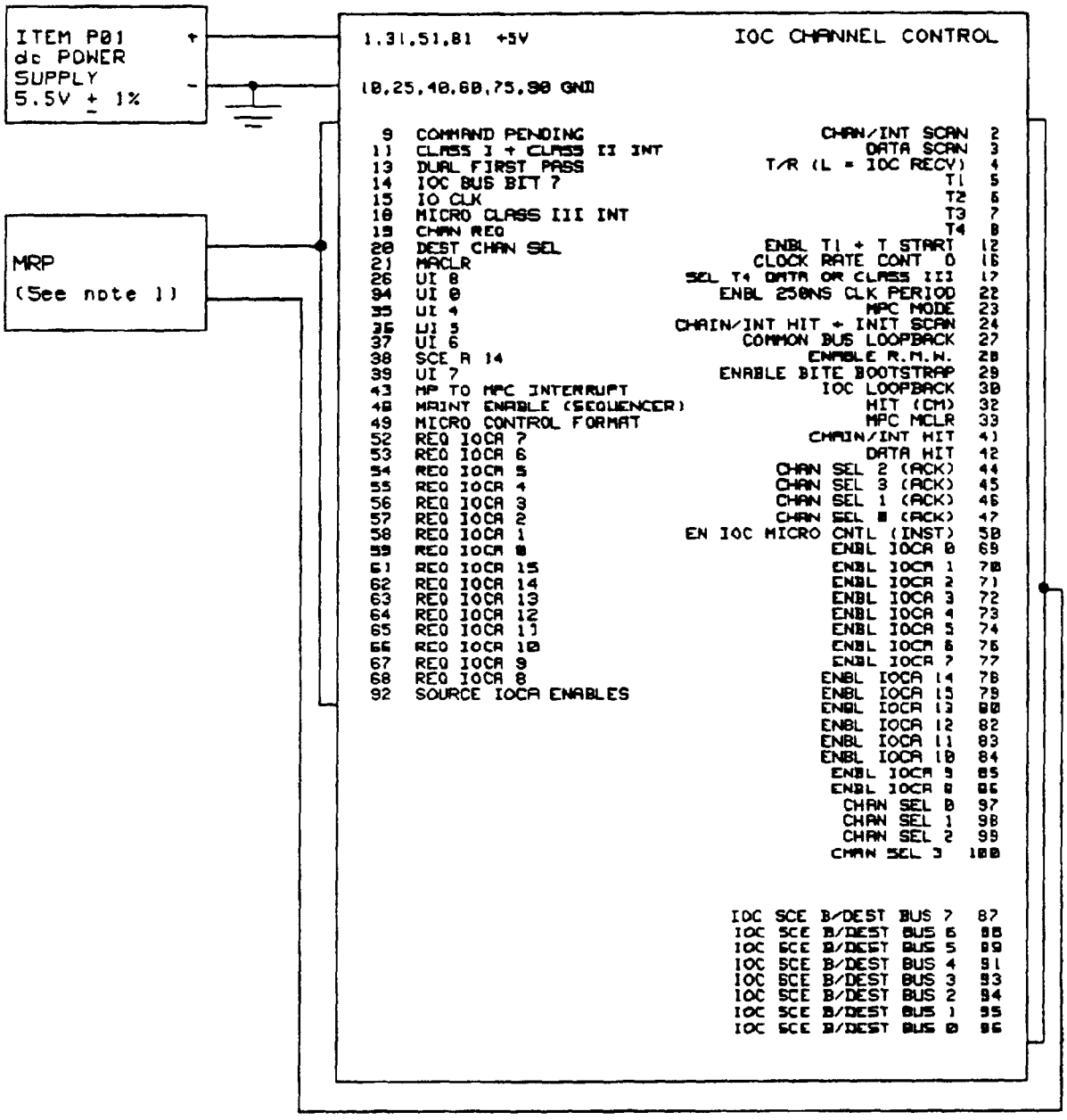
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AWH operating temperature and component temperature test circuit.

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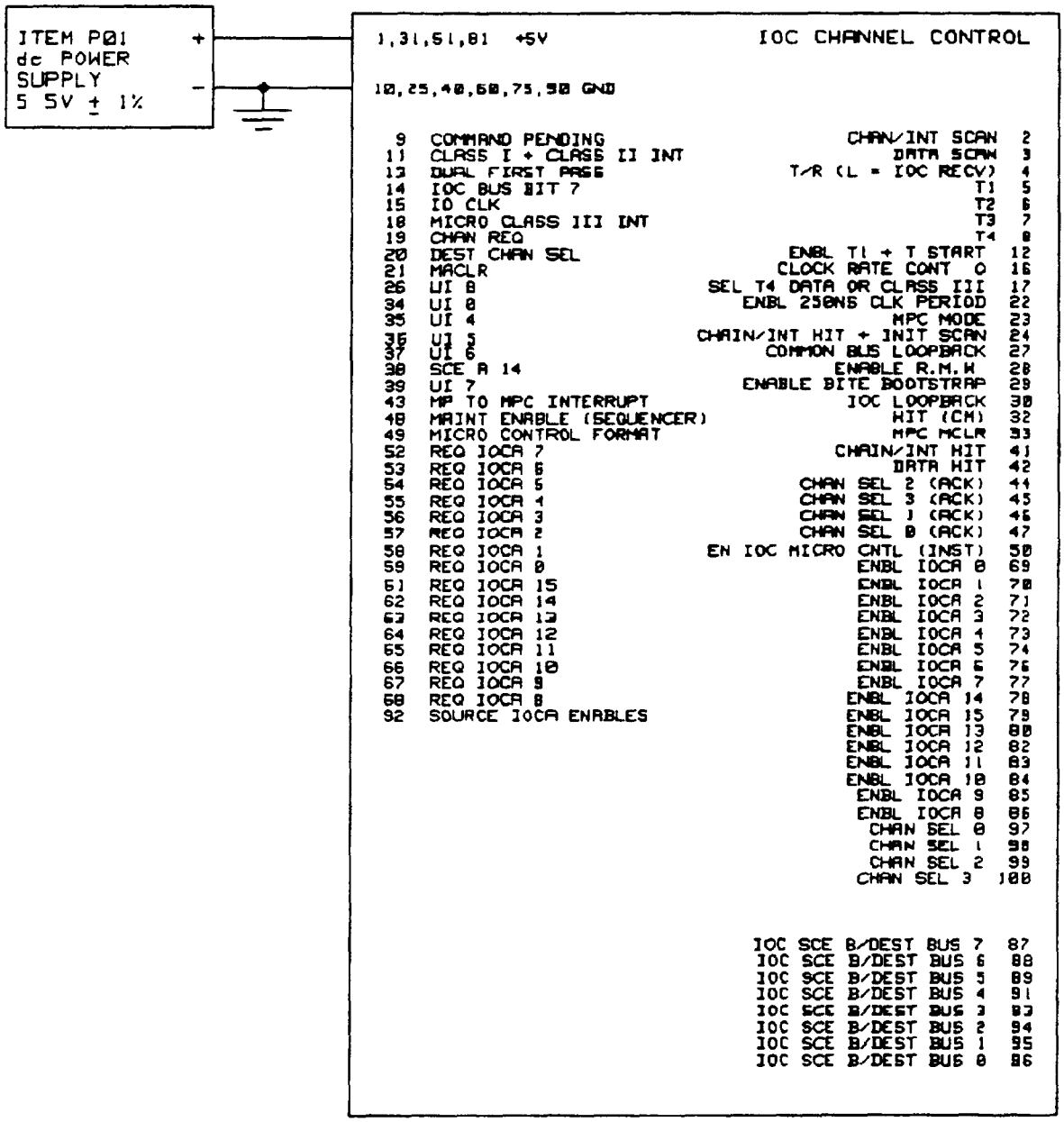
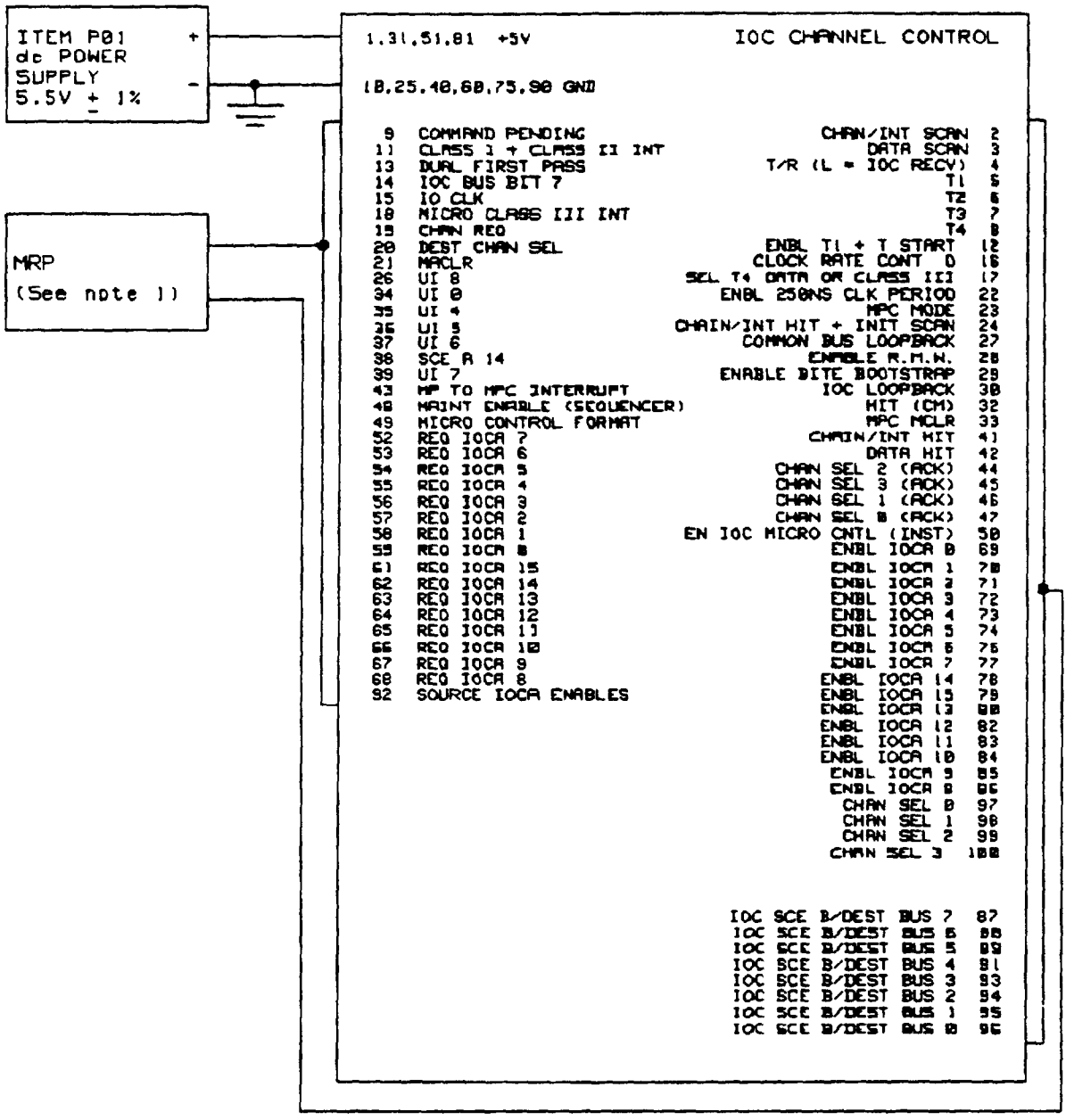


FIGURE 4. AWH vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AWH life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.85 W (max); 2.58 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
COMM PEND	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CLASS I+C	-1.2	0.4	2.4	-0.0	0.02	0.1	125
DUAL FIRST	-1.2	0.4	2.4	-0.6	0.02	0.1	75
BUS BIT 7	-1.2	0.4	2.4	-2.0	0.05	1.0	50
IO CLOCK	-1.2	0.4	2.4	-10.0	0.25	5.0	100
MICRO CIII	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CHAN REQ	-1.2	0.4	2.4	-1.2	0.04	0.2	75
DEST CHAN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MACLR	-1.2	0.4	2.4	-9.4	0.12	0.6	150
UI 8	-1.2	0.4	2.4	-1.2	0.04	0.2	75
UI 0	-1.5	0.4	2.4	-0.6	0.04	0.2	75
UI 4	-1.5	0.4	2.4	-0.8	0.04	0.2	75
UI 5	-1.5	0.4	2.4	-0.8	0.04	0.2	75
UI 6	-1.5	0.4	2.4	-0.8	0.04	0.2	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SCE A 14	-1.2	0.4	2.4	-0.6	0.02	0.1	75
UI 7	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP TO MPC	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MAINT ENBL	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MICRO CONT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
REQ IOCA 7	NA	0.4	2.4	-1.26	-0.54	0.12	50
REQ IOCA 6	NA	0.4	2.4	-1.26	-0.54	0.12	75
REQ IOCA 5	NA	0.4	2.4	-1.26	-0.54	0.12	50
REQ IOCA 4	NA	0.4	2.4	-1.26	-0.54	0.12	75
REQ IOCA 3	NA	0.4	2.4	-1.26	-0.54	0.12	75
REQ IOCA 2	NA	0.4	2.4	-1.26	-0.54	0.12	75
REQ IOCA 1	NA	0.4	2.4	-1.26	-0.54	0.12	50
REQ IOCA 0	NA	0.4	2.4	-1.26	-0.54	0.12	50
REQ IOCA 15	NA	0.4	2.4	-1.26	-0.54	0.12	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
REQ IOCA 14	NA	0.4	2.4	-1.26	-0.54	0.12	50
REQ IOCA 13	NA	0.4	2.4	-1.26	-0.54	0.12	50
REQ IOCA 12	NA	0.4	2.4	-1.26	-0.54	0.12	50
REQ IOCA 11	NA	0.4	2.4	-1.26	-0.54	0.12	50
REQ IOCA 10	NA	0.4	2.4	-1.26	-0.54	0.12	75
REQ IOCA 9	NA	0.4	2.4	-1.26	-0.54	0.12	75
REQ IOCA 8	NA	0.4	2.4	-1.26	-0.54	0.12	75
SOURCE IOC	-1.2	0.4	2.4	-1.2	0.04	0.2	75
BUS 3	NA	0.4	2.4	-0.25	0.07	NA	75
BUS 2	NA	0.4	2.4	-0.25	0.07	NA	75
BUS 1	NA	0.4	2.4	-0.25	0.07	NA	75
BUS 0	NA	0.4	2.4	-0.25	0.07	NA	75
BUS 7	NA	0.4	2.4	-0.25	0.07	NA	75
BUS 6	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
BUS 5	NA	0.4	2.4	-0.25	0.07	NA	75
BUS 4	NA	0.4	2.4	-0.25	0.07	NA	75

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
CHAN/INT	2.4	0.5	-0.4	6.0	-100.0	NA
DATA SCAN	2.4	0.5	-0.4	6.0	-100.0	NA
T/R	2.4	0.5	-0.4	6.0	-100.0	NA
T1	2.4	0.5	-0.4	6.0	-100.0	NA
T2	2.4	0.5	-0.4	6.0	-100.0	NA
T3	2.4	0.5	-0.4	6.0	-100.0	NA
T4	2.4	0.5	-0.4	6.0	-100.0	NA
ENBL T1	2.4	0.5	-0.4	6.0	-60.0	NA
CLOCK RATE	2.4	0.5	-0.4	6.0	-60.0	NA
SEL T4	2.4	0.5	-0.16	1.6	-15.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
ENBL 250NS	2.4	0.5	-0.4	6.0	-60.0	NA
MPC MODE	2.4	0.5	-0.4	6.0	-100.0	NA
INIT SCAN	2.4	0.5	-0.4	6.0	-60.0	NA
COMMON BUS	2.4	0.5	-0.16	1.6	-15.0	NA
ENABLE RMW	2.4	0.5	-0.16	1.6	-15.0	NA
ENBL BITE	2.4	0.5	-0.16	1.6	-15.0	NA
IOC LPBACK	2.4	0.5	-0.16	1.6	-15.0	NA
HIT (CM)	2.4	0.5	-0.4	6.0	-60.0	NA
MPC MCLR	2.4	0.5	-0.16	1.6	-30.0	NA
CHAIN/INT	2.4	0.5	-0.16	1.6	-20.0	NA
DATA HIT	2.4	0.5	-0.16	1.6	-20.0	NA
CHAN SEL 2	2.4	0.5	-0.16	1.6	-20.0	NA
CHAN SEL 3	2.4	0.5	-0.16	1.6	-20.0	NA
CHAN SEL 1	2.4	0.5	-0.16	1.6	-20.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
CHAN SEL 0	2.4	0.5	-0.16	1.6	-20.0	NA
EN IOCA MIC	2.4	0.5	-0.4	6.0	-40.0	NA
EN IOCA 0	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 1	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 2	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 3	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 4	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 5	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 6	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 7	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 14	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 15	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 13	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 12	2.4	0.5	-0.4	6.0	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
EN IOCA 11	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 10	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 9	2.4	0.5	-0.4	6.0	-60.0	NA
EN IOCA 8	2.4	0.5	-0.4	6.0	-60.0	NA
BUS 7	2.4	0.5	-0.4	6.0	-60.0	75
BUS 6	2.4	0.5	-0.4	6.0	-60.0	75
BUS 5	2.4	0.5	-0.4	6.0	-60.0	75
BUS 4	2.4	0.5	-0.4	6.0	-60.0	75
BUS 3	2.4	0.5	-0.4	6.0	-60.0	75
BUS 2	2.4	0.5	-0.4	6.0	-60.0	75
BUS 1	2.4	0.5	-0.4	6.0	-60.0	75
BUS 0	2.4	0.5	-0.4	6.0	-60.0	75
CHAN SEL 0	2.4	0.5	-0.16	1.6	-20.0	NA
CHAN SEL 1	2.4	0.5	-0.16	1.6	-20.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
CHAN SEL 2	2.4	0.5	-0.16	1.6	-20.0	NA
CHAN SEL 3	2.4	0.5	-0.16	1.6	-20.0	NA

6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	BUS BIT 7 to IO CLOCK	23	23	23	-	-	-	nS	12
2	th	IO CLOCK to BUS BIT 7	11	11	11	-	-	-	nS	12
3	tsu	MP TO MPC to IO CLOCK	88	88	88	-	-	-	nS	12
4	th	IO CLOCK to MP TO MPC	4	4	4	-	-	-	nS	12
5	tsu	REQ IOCA 0 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 1 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 2 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 3 to IO CLOCK	6	6	6	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	REQ IOCA 4 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 5 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 6 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 7 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 8 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 9 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 10 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 11 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 12 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 13 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 14 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 15 to IO CLOCK	6	6	6	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	th	IO CLOCK to REQ IOCA 0	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 1	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 2	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 3	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 4	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 5	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 6	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 7	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 8	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 9	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 10	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 11	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 12	43	43	43	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	th	IO CLOCK to REQ IOCA 13	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 14	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 15	43	43	43	-	-	-	nS	12
7	tsu	COMM PEND to IO CLOCK	12	12	12	-	-	-	nS	12
8	th	IO CLOCK to COMM PEND	14	14	14	-	-	-	nS	12
9	tsu	CLASS I+C to IO CLOCK	23	23	23	-	-	-	nS	12
10	th	IO CLOCK to CLASS I+C	6	6	6	-	-	-	nS	12
11	tpd	IO CLOCK to T1	-	-	-	64	64	64	nS	12
12	tpd	IO CLOCK to EN IOCA 0	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 1	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 2	-	-	-	96	96	96	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	IO CLOCK to EN IOCA 3	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 4	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 5	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 6	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 7	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 8	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 9	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 10	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 11	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 12	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 13	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 14	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 15	-	-	-	96	96	96	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
13	tpd	IO CLOCK to T/R	-	-	-	90	90	90	nS	12
14	tpd	IO CLOCK to MPC MODE	-	-	-	64	64	64	nS	12
15	tpd	UI 0 to SEL T4	-	-	-	58	58	58	nS	12
15	tpd	UI 4 to SEL T4	-	-	-	58	58	58	nS	12
15	tpd	UI 5 to SEL T4	-	-	-	58	58	58	nS	12
15	tpd	UI 6 to SEL T4	-	-	-	58	58	58	nS	12
16	tpd	IO CLOCK to DATA SCAN	-	-	-	79	79	79	nS	12
16	tpd	IO CLOCK to CHAN/INT	-	-	-	79	79	79	nS	12
17	tpd	SCE A 14 to EN IOC MIC	-	-	-	37	37	37	nS	12

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6.4 Module function. The IOC channel control module contains a request encoder, bus scan timing circuits, select-acknowledge control circuits, and an enable generator. The IOC channel select module is responsible for receiving the Input/Output Channel Adapter (IOCA) requests, prioritizing them, and enabling the priority IOCA (channel F having the highest priority). Additionally, the IOC channel select module is responsible for generating data and chain/interrupt scans and directing them to the IOCA channel control module. The request encoder receives the IOCA requests from the IOCA channel control module (the highest numbered channel having priority), encodes them, and sends channel select/enable data to the IOC control memory module and the IOCA channel control module. The bus scan timing circuits generate data and chain/interrupt scans to the IOCA channel control module resulting in a search for a data "hit" or a chain/interrupt "hit" from a channel. Select/acknowledge control circuits latch the channel select bits from the select encoder, delay them to provide acknowledges to requests, and send the channel select bits to IOC control memory. The enable generator is composed of a decoder to decode the channel select bits and a multiplexer to place the enables onto the source B/destination bus.

6.4.1 Channel request latch and priority. The channel request latch receives inputs from the input/output channel adapters (req IOCA 15-0). IOCA requests are latched in at T1 with a gated Input/Output (I/O) clock and are sent to an IOCA select encoder circuit. This circuit is composed of two 8-line to 3-line encoders (always enabled) that reduce the 16 request inputs to six lines and apply them to the channel select multiplexer. Enable outputs from the encoder circuits are ORed together to form the IOCA hit signal that starts the chain/interrupt scan. Enable output from the encoder for IOCA bits 15-8 is the select bit for the channel select multiplexer circuit. Outputs from the channel select multiplexer are placed on the channel select bus bits 3-0.

6.4.2 IOCA enable decoder. The IOCA enable decoder receives inputs from the channel select bus and sends IOCA enable bits to the IOCA modules. Enables for the decoder are channel select bus bit 3 and the enable IOCA select signal. Outputs are also sent to the IOCA enable multiplexer which receives enables from the IOC sequencer module (source IOCA enables, select enable 0-7). The IOCA enable multiplexer places enable information onto the source B/destination bus.

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6.4.3 IOC bus scan timing circuits. The IOC bus scan timing circuits are responsible for generating the following signals:

(a) Chain/interrupt scan.

(b) Data scan.

(c) IOC transmit/receive.

For a chain operation, the IOC enables chain or macro interrupt scans during T1 time by setting the chain/interrupt scan. Also during T1, the IOCA presents a prioritized request for chaining activity - the IOC selecting the highest priority IOCA during T2 and T3 to enable IOCA data during T3 and T4. The IOCA then selects the type of chain request during T3 time by using bits of the IOC data bus. The IOC sequencer (BWAA), IOC arithmetic logic unit (BWE), and IOC control memory (AWF) accomplish all actions required to fetch and execute the chain instructions. The IOC, during execution of the chain, does the following:

(a) Disables any further scanning for data or chain requests.

(b) Changes the IOC clock from a 150 nS cycle to a 250 nS cycle time.

(c) Initiates IOC data bus activity and timing as required to perform the chain instructions.

For a class III interrupt transaction, the IOC will enable chain or macro interrupt scans during T1 time by setting the chain/ interrupt scan. The IOCA presents its prioritized request and the IOC will select the highest priority IOCA during T2 and T3. Actions required to generate a class III interrupt to the CP are done by the IOC during T1 of the next bus cycle. The IOC scans IOCA data requests on alternate bus cycles. The occurrence of a data "hit" results in another data scan with the IOC continuing data scans until no more data requests are present. Data scans are enabled during T1 time with the IOCAs presenting their prioritized requests to the IOC. The highest priority IOCA is enabled at T2/T3 enabling the IOCA to specify the type of transfer during T3 and T4. The remaining bus cycles complete the data transaction.

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6.4.4 Channel/Micro Program Controller (MPC) select multiplexer. The channel/MPC select multiplexer selects either channel information from the channel select register or channel information from the IOCA enable multiplexer (data sample of source B/destination bus bits 7-0) for application to the select acknowledge multiplexer. Enable for the channel/MPC select multiplexer is the MPC mode signal from the MPC mode flip-flop.

6.4.5 Select/acknowledge multiplexer. The select/acknowledge multiplexer selects either outputs from the channel/MPC select multiplexer or channel select/acknowledge bits from the select/acknowledge control circuits. Enables are the select IOC channel acknowledge signal from the IOC bus scan timing circuits, and the select function performed by the MPC mode or IOC receive signal. Outputs from the select/acknowledge multiplexer are wired ORed with the outputs from the channel select multiplexer for application to the channel select bus.

6.4.6 Select/acknowledge control circuits. The select/acknowledge control circuits receive the channel select information from the channel/select multiplexer or the select acknowledge multiplexer and delay the channel select data to be supplied as an acknowledge on the channel select bus.

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6.4.7 Description of I/O pins.

Signal name	Pin number	I/O	Description
COMM PEND	9	I	COMM PENDING
CLASS I+C	11	I	CLASS I + CLASS II INT
DUAL FIRST	13	I	DUAL FIRST PASS
BUS BIT 7	14	I	IOC BUS BIT 7
IO CLOCK	15	I	IO CLK
MICRO CIII	18	I	MICRO CLASS III INT
CHAN REQ	19	I	CHAN REQ
DEST CHAN	20	I	DEST CHAN SEL
MACLR	21	I	MACLR
UI 8	26	I	UI 8
UI 0	34	I	UI 0
UI 4	35	I	UI 4
UI 5	36	I	UI 5
UI 6	37	I	UI 6
SCE A 14	38	I	SCE A 14
UI 7	39	I	UI 7
MP TO MPC	43	I	MP TO MPC INTERRUPT
MAINT ENBL	48	I	MAINT ENABLE (SEQUENCER)
MICRO CONT	49	I	MICRO CONTROL FORMAT
REQ IOCA 7	52	I	REQ IOCA 7

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
REQ IOCA 6	53	I	REQ IOCA 6
REQ IOCA 5	54	I	REQ IOCA 5
REQ IOCA 4	55	I	REQ IOCA 4
REQ IOCA 3	56	I	REQ IOCA 3
REQ IOCA 2	57	I	REQ IOCA 2
REQ IOCA 1	58	I	REQ IOCA 1
REQ IOCA 0	59	I	REQ IOCA 0
REQ IOCA 15	61	I	REQ IOCA 15
REQ IOCA 14	62	I	REQ IOCA 14
REQ IOCA 13	63	I	REQ IOCA 13
REQ IOCA 12	64	I	REQ IOCA 12
REQ IOCA 11	65	I	REQ IOCA 11
REQ IOCA 10	66	I	REQ IOCA 10
REQ IOCA 9	67	I	REQ IOCA 9
REQ IOCA 8	68	I	REQ IOCA 8
CHAN/INT	2	O	CHAN/INT SCAN
DATA SCAN	3	O	DATA SCAN
T/R	4	O	T/R (L = IOC RECV)
T1	5	O	T1
T2	6	O	T2

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
T3	7	O	T3
T4	8	O	T4
ENBL T1	12	O	ENBL T1 + T START
CLOCK RATE	16	O	CLOCK RATE CONT 0
SEL T4	17	O	SEL T4 DATA OR CLASS III
ENBL 250NS	22	O	ENBL 250NS CLK PERIOD
MPC MODE	23	O	MPC MODE
INIT SCAN	24	O	CHAIN/INT HIT + INIT SCAN
COMMON BUS	27	O	COMMON BUS LOOPBACK
ENABLE RMW	28	O	ENABLE R. M. W.
ENBL BITE	29	O	ENABLE BITE BOOTSTRAP
IOC LPBACK	30	O	IOC LOOPBACK
HIT (CM)	32	O	HIT (CM)
MPC MCLR	33	O	MPC MCLR
CHAIN/INT	41	O	CHAIN/INT HIT
DATA HIT	42	O	DATA HIT
CHAN SEL 2	44	O	CHAN SEL 2 (ACK)
CHAN SEL 3	45	O	CHAN SEL 3 (ACK)
CHAN SEL 1	46	O	CHAN SEL 1 (ACK)
CHAN SEL 0	47	O	CHAN SEL 0 (ACK)

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
EN IOC MIC	50	O	EN IOC MICRO CNTL (INST)
EN IOCA 0	69	O	ENBL IOCA 0
EN IOCA 1	70	O	ENBL IOCA 1
EN IOCA 2	71	O	ENBL IOCA 2
EN IOCA 3	72	O	ENBL IOCA 3
EN IOCA 4	73	O	ENBL IOCA 4
EN IOCA 5	74	O	ENBL IOCA 5
EN IOCA 6	76	O	ENBL IOCA 6
EN IOCA 7	77	O	ENBL IOCA 7
EN IOCA 14	78	O	ENBL IOCA 14
EN IOCA 15	79	O	ENBL IOCA 15
EN IOCA 13	80	O	ENBL IOCA 13
EN IOCA 12	82	O	ENBL IOCA 12
EN IOCA 11	83	O	ENBL IOCA 11
EN IOCA 10	84	O	ENBL IOCA 10
EN IOCA 9	85	O	ENBL IOCA 9
EN IOCA 8	86	O	ENBL IOCA 8
BUS 7	87	I/O	IOC SCE B/DEST BUS 7
BUS 6	88	I/O	IOC SCE B/DEST BUS 6
BUS 5	89	I/O	IOC SCE B/DEST BUS 5

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
BUS 4	91	I/O	IOC SCE B/DEST BUS 4
SOURCE IOC	92	I	SOURCE IOCA ENABLES
BUS 3	93	I/O	IOC SCE B/DEST BUS 3
BUS 2	94	I/O	IOC SCE B/DEST BUS 2
BUS 1	95	I/O	IOC SCE B/DEST BUS 1
BUS 0	96	I/O	IOC SCE B/DEST BUS 0
CHAN SEL 0	97	O	CHAN SEL 0
CHAN SEL 1	98	O	CHAN SEL 1
CHAN SEL 2	99	O	CHAN SEL 2
CHAN SEL 3	100	O	CHAN SEL 3

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N721)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to	B to	Min	Max		
A	Isolation, Frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal, and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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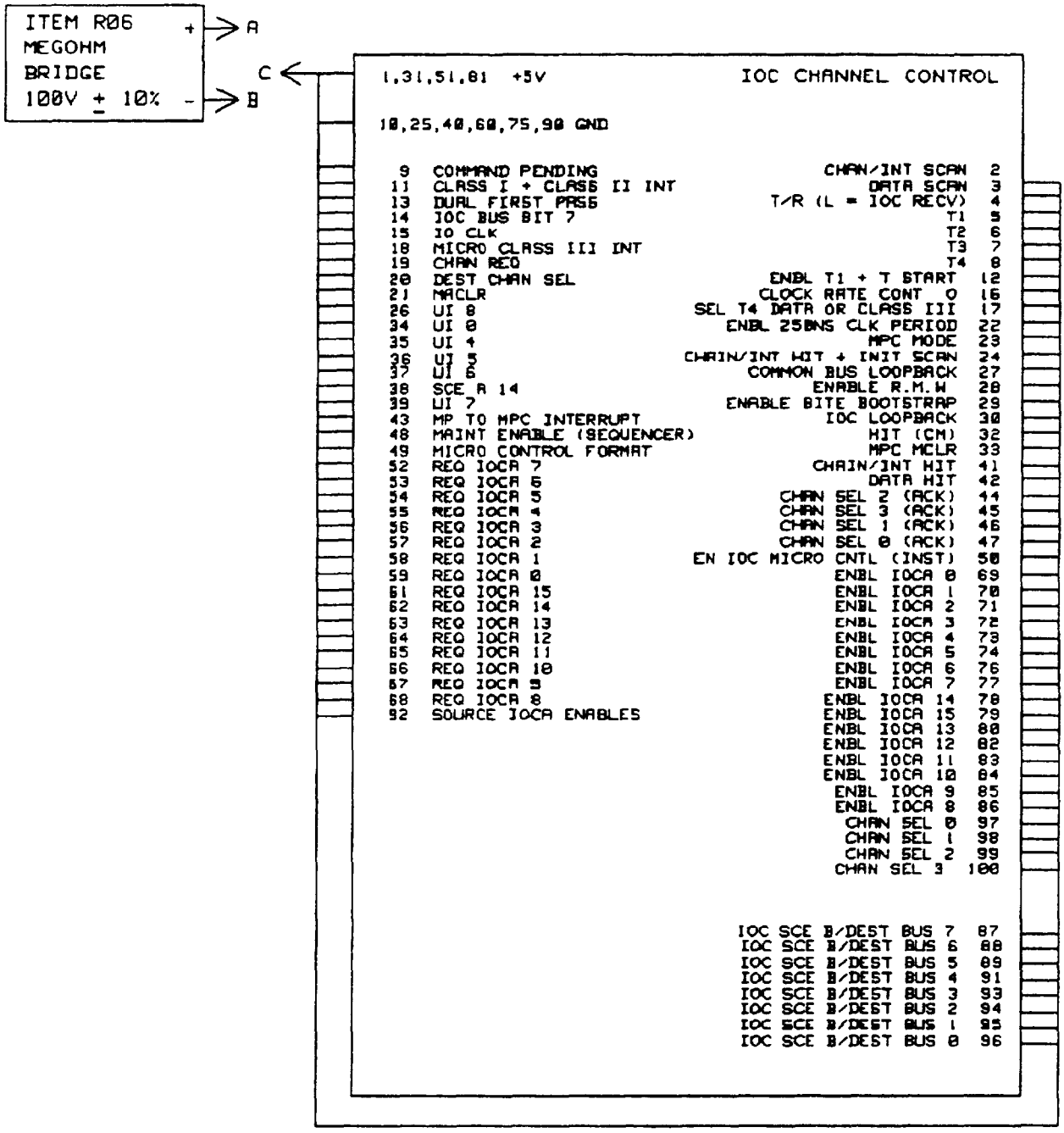


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, +5V	1	31				
B6	Continuity, +5V	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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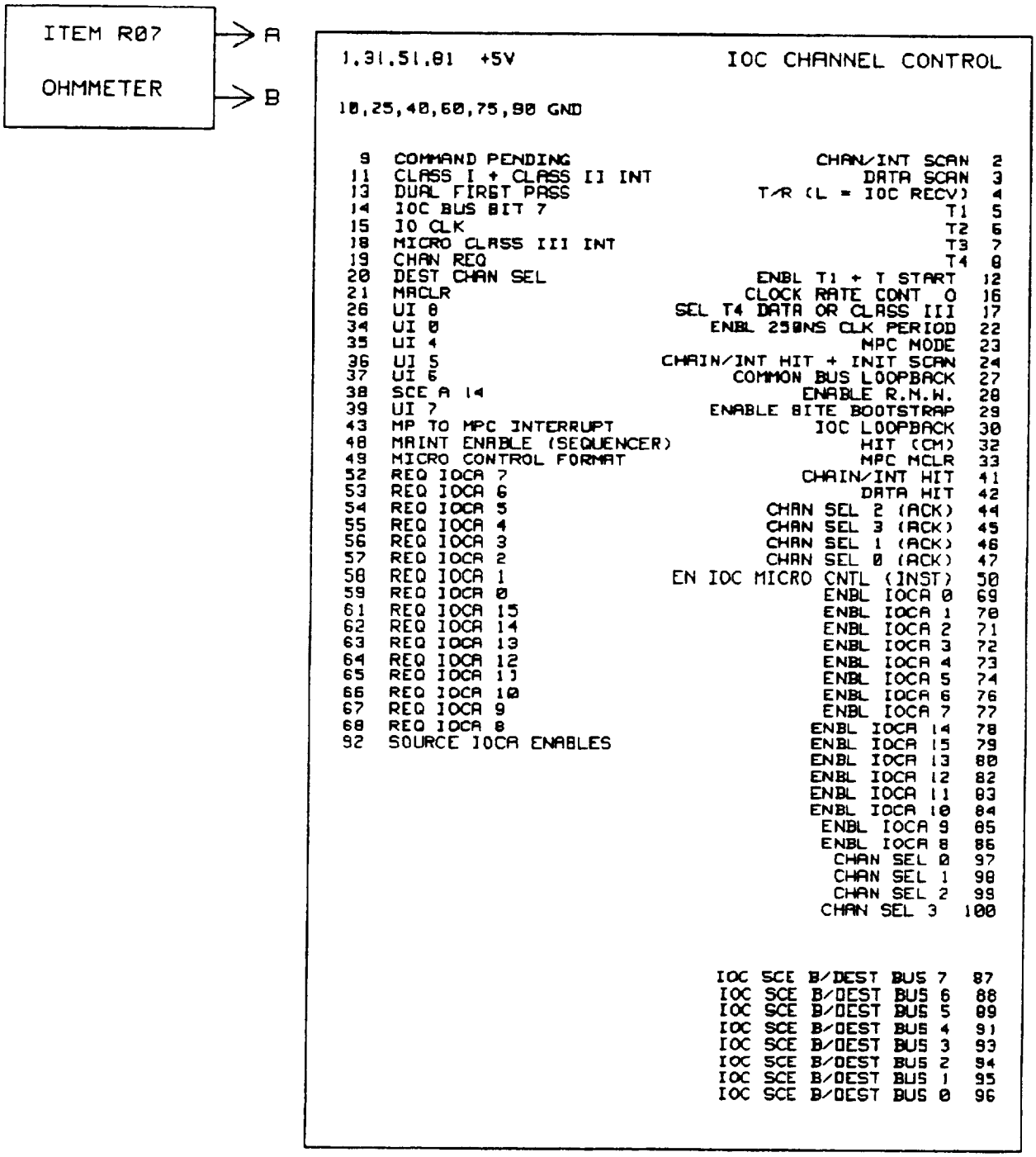


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1, 31, 51, 81	10, 25, 40, 60, 75, 90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV, -70 mV	C
Power supply current limit	1.26 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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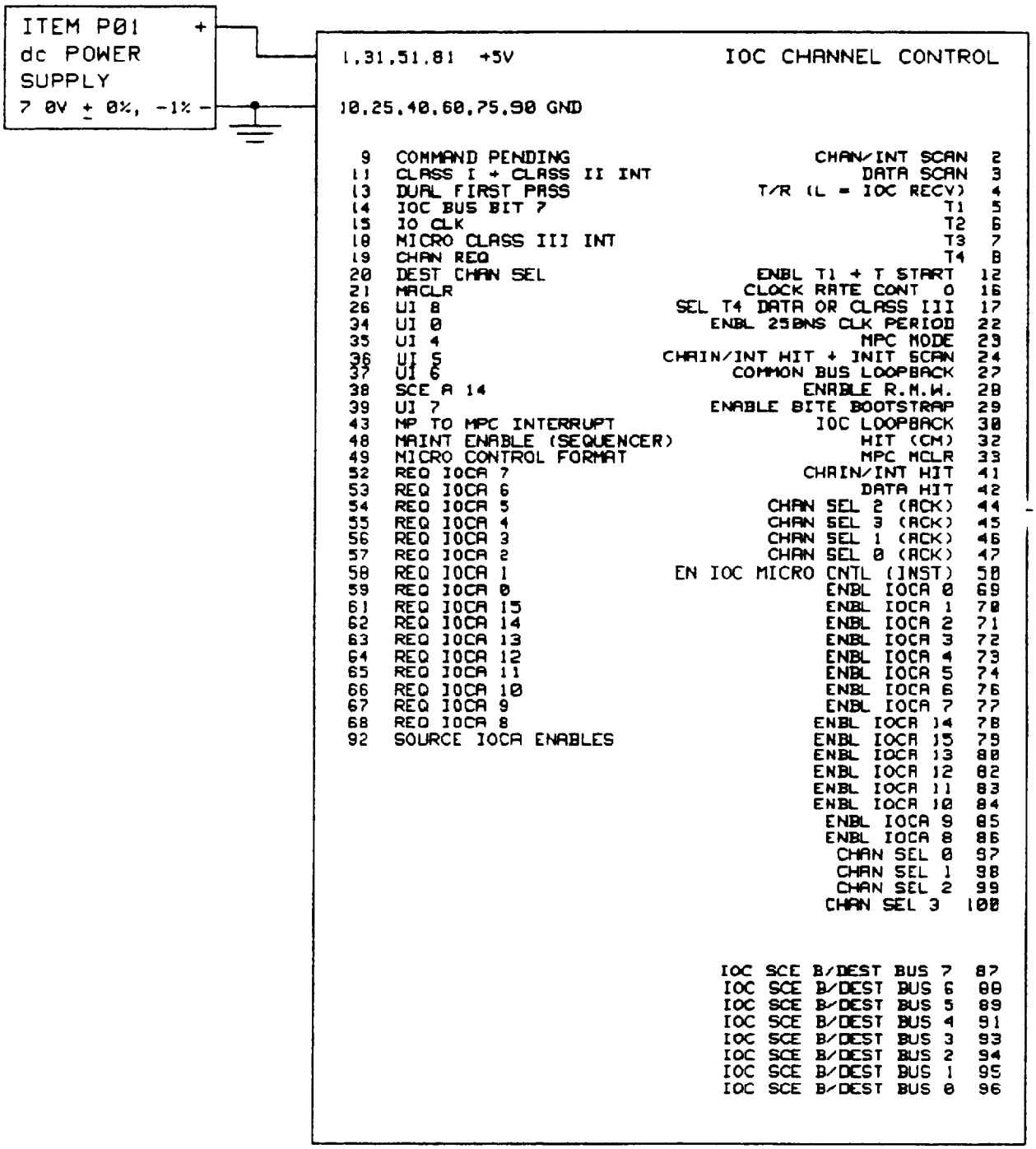


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	COMM PEND	9	5.5	0.0	0.1	mA	1-3
D2	CLASS I+C	11					
D3	DUAL FIRST	13	5.5	0.0	0.1	mA	1-3
D4	BUS BIT 7	14	5.5	0.0	1.0	mA	1-3
D5	IO CLOCK	15	5.5	0.0	5.0	mA	1-3
D6	MICRO CIII	18	5.5	0.0	0.1	mA	1-3
D7	CHAN REQ	19	5.5	0.0	0.2	mA	1-3
D8	DEST CHAN	20	5.5	0.0	0.1	mA	1-3
D9	MACLR	21	5.5	0.0	0.6	mA	1-3
D10	UI 8	26	5.5	0.0	0.2	mA	1-3
D11	UI 0	34					
D12	UI 4	35					
D13	UI 5	36					
D14	UI 6	37	5.5	0.0	0.2	mA	1-3
D15	SCE A 14	38	5.5	0.0	0.1	mA	1-3
D16	UI 7	39					
D17	MP TO MPC	43					
D18	MAINT ENBL	48					
D19	MICRO CONT	49	5.5	0.0	0.1	mA	1-3
D20	REQ IOCA 7	52	5.5	-0.02	0.12	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	REQ IOCA 6	53	5.5	-0.02	0.12	mA	1-3
D22	REQ IOCA 5	54					
D23	REQ IOCA 4	55					
D24	REQ IOCA 3	56					
D25	REQ IOCA 2	57					
D26	REQ IOCA 1	58					
D27	REQ IOCA 0	59					
D28	REQ IOCA 15	61					
D29	REQ IOCA 14	62					
D30	REQ IOCA 13	63					
D31	REQ IOCA 12	64					
D32	REQ IOCA 11	65					
D33	REQ IOCA 10	66					
D34	REQ IOCA 9	67					
D35	REQ IOCA 8	68	5.5	-0.02	0.12	mA	1-3
D36	SOURCE IOC	92	5.5	0.0	0.2	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D36
Power supply current limit	1.05 A	NA	D1-D36
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D36
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D36
low, (VL)	0.0 V	± 100 mV	D1-D36
Input current measurement	NA	± 5 uA	D1-D3
	NA	± 50 uA	D4
	NA	± 250 uA	D5
	NA	± 5 uA	D6
	NA	± 10 uA	D7
	NA	± 5 uA	D8
	NA	± 30 uA	D9
	NA	± 10 uA	D10-D14
	NA	± 5 uA	D15-D19
	NA	± 7 uA	D20-D35
	NA	± 10 uA	D36

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	CHAN/INT	2	0.0	-325.0	-100.0	mA	1-4
E2	DATA SCAN	3					
E3	T/R	4					
E4	T1	5					
E5	T2	6					
E6	T3	7					
E7	T4	8	0.0	-325.0	-100.0	mA	1-4
E8	ENBL T1	12	0.0	-150.0	-60.0	mA	1-4
E9	CLOCK RATE	16	0.0	-150.0	-60.0	mA	1-4
E10	SEL T4	17	0.0	-100.0	-15.0	mA	1-4
E11	ENBL 250NS	22	0.0	-150.0	-60.0	mA	1-4
E12	MPC MODE	23	0.0	-325.0	-100.0	mA	1-4
E13	INIT SCAN	24	0.0	-150.0	-60.0	mA	1-4
E14	COMMON BUS	27	0.0	-100.0	-15.0	mA	1-4
E15	ENABLE RMW	28					
E16	ENBL BITE	29					
E17	IOC LPBACK	30	0.0	-100.0	-15.0	mA	1-4
E18	HIT (CM)	32	0.0	-150.0	-60.0	mA	1-4
E19	MPC MCLR	33	0.0	-260.0	-30.0	mA	1-4
E20	CHAIN/INT	41	0.0	-100.0	-20.0	mA	1-4
E21	DATA HIT	42	0.0	-100.0	-20.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	CHAN SEL 2	44	0.0	-100.0	-20.0	mA	1-4
E23	CHAN SEL 3	45					
E24	CHAN SEL 1	46					
E25	CHAN SEL 0	47	0.0	-100.0	-20.0	mA	1-4
E26	EN IOC MIC	50	0.0	-100.0	-40.0	mA	1-4
E27	EN IOCA 0	69	0.0	-150.0	-60.0	mA	1-4
E28	EN IOCA 1	70					
E29	EN IOCA 2	71					
E30	EN IOCA 3	72					
E31	EN IOCA 4	73					
E32	EN IOCA 5	74					
E33	EN IOCA 6	76					
E34	EN IOCA 7	77					
E35	EN IOCA 14	78					
E36	EN IOCA 15	79					
E37	EN IOCA 13	80					
E38	EN IOCA 12	82					
E39	EN IOCA 11	83					
E40	EN IOCA 10	84					
E41	EN IOCA 9	85					
E42	EN IOCA 8	86	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	BUS 7	87	0.0	-150.0	-60.0	mA	1-4
E44	BUS 6	88					
E45	BUS 5	89					
E46	BUS 4	91					
E47	BUS 3	93					
E48	BUS 2	94					
E49	BUS 1	95					
E50	BUS 0	96	0.0	-150.0	-60.0	mA	1-4
E51	CHAN SEL 0	97	0.0	-100.0	-20.0	mA	1-4
E52	CHAN SEL 1	98					
E53	CHAN SEL 2	99					
E54	CHAN SEL 3	100	0.0	-100.0	-20.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E54
Power supply current limit	1.05 A	NA	E1-E54

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E54
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E54 E1-E54
Output current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA	± 11.2 mA ± 4.50 mA ± 4.25 mA ± 4.50 mA ± 11.2 mA ± 4.50 mA ± 4.25 mA ± 4.50 mA ± 11.5 mA ± 4.0 mA ± 3.0 mA ± 4.50 mA ± 4.0 mA	E1-E7 E8,E9 E10 E11 E12 E13 E14-E17 E18 E19 E20-E25 E26 E27-E50 E51-E54

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	COMM PEND	9	2.4	-0.2	0.02	mA	1-3
F2	CLASS I+C	11	2.4	-0.6	-0.02	mA	1-3
F3	DUAL FIRST	13	2.4	-0.6	-0.02	mA	1-3
F4	BUS BIT 7	14	2.4	-2.0	0.05	mA	1-3
F5	IO CLOCK	15	2.4	-10.0	0.25	mA	1-3
F6	MICRO CIII	18	2.4	-0.6	0.02	mA	1-3
F7	CHAN REQ	19	2.4	-1.2	0.04	mA	1-3
F8	DEST CHAN	20	2.4	-0.6	0.02	mA	1-3
F9	MACLR	21	2.4	-9.4	0.12	mA	1-3
F10	UI 8	26	2.4	-1.2	0.04	mA	1-3
F11	UI 0	34	2.4	-0.6	0.04	mA	1-3
F12	UI 4	35	2.4	-0.8	0.04	mA	1-3
F13	UI 5	36					
F14	UI 6	37	2.4	-0.8	0.04	mA	1-3
F15	SCE A 14	38	2.4	-0.6	0.02	mA	1-3
F16	UI 7	39	2.4	-0.2	0.02	mA	1-3
F17	MP TO MPC	43	2.4	-0.6	0.02	mA	1-3
F18	MAINT ENBL	48	2.4	-0.2	0.02	mA	1-3
F19	MICRO CONT	49	2.4	-0.6	0.02	mA	1-3
F20	REQ IOCA 7	52	2.4	-1.26	-0.54	mA	1-3
F21	REQ IOCA 6	53	2.4	-1.26	-0.54	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	REQ IOCA 5	54	2.4	-1.26	-0.54	mA	1-3
F23	REQ IOCA 4	55					
F24	REQ IOCA 3	56					
F25	REQ IOCA 2	57					
F26	REQ IOCA 1	58					
F27	REQ IOCA 0	59					
F28	REQ IOCA 15	61					
F29	REQ IOCA 14	62					
F30	REQ IOCA 13	63					
F31	REQ IOCA 12	64					
F32	REQ IOCA 11	65					
F33	REQ IOCA 10	66					
F34	REQ IOCA 9	67					
F35	REQ IOCA 8	68	2.4	-1.26	-0.54	mA	1-3
F36	BUS 7	87	2.4	-0.25	0.07	mA	1-3
F37	BUS 6	88					
F38	BUS 5	89					
F39	BUS 4	91	2.4	-0.25	0.07	mA	1-3
F40	SOURCE IOC	92	2.4	-1.2	0.04	mA	1-3
F41	BUS 3	93	2.4	-0.25	0.07	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F42	BUS 2	94	2.4	-0.25	0.07	mA	1-3
F43	BUS 1	95					
F44	BUS 0	96	2.4	-0.25	0.07	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F44
Power supply current limit	1.05 A	NA	F1-F44
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F44
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F44
low, (VL)	0.0 V	± 100 mV	F1-F44
Input current measurement	NA	± 11 uA	F1
	NA	± 29 uA	F2,F3
	NA	± 102 uA	F4
	NA	± 510 uA	F5
	NA	± 31 uA	F6
	NA	± 62 uA	F7
	NA	± 31 uA	F8
	NA	± 480 uA	F9

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 62 uA	F10
	NA	± 32 uA	F11
	NA	± 42 uA	F12-F14
	NA	± 31 uA	F15
	NA	± 11 uA	F16
	NA	± 31 uA	F17
	NA	± 11 uA	F18
	NA	± 31 uA	F19
	NA	± 36 uA	F20-F35
	NA	± 16 uA	F36-F39
	NA	± 62 uA	F40
	NA	± 16 uA	F41-F44

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	COMM PEND	9	0.4	-0.2	0.02	mA	1-3
G2	CLASS I+C	11	0.4	-0.6	0.02	mA	1-3
G3	DUAL FIRST	13	0.4	-0.6	0.02	mA	1-3
G4	BUS BIT 7	14	0.4	-2.0	0.05	mA	1-3
G5	IO CLOCK	15	0.4	-10.0	0.25	mA	1-3
G6	MICRO CIII	18	0.4	-0.6	0.02	mA	1-3
G7	CHAN REQ	19	0.4	-1.2	0.04	mA	1-3
G8	DEST CHAN	20	0.4	-0.6	0.02	mA	1-3
G9	MACLR	21	0.4	-9.4	0.12	mA	1-3
G10	UI 8	26	0.4	-1.2	0.04	mA	1-3
G11	UI 0	34	0.4	-0.6	0.04	mA	1-3
G12	UI 4	35	0.4	-0.8	0.04	mA	1-3
G13	UI 5	36	|	|	|	|	|
G14	UI 6	37	0.4	-0.8	0.04	mA	1-3
G15	SCE A 14	38	0.4	-0.6	0.02	mA	1-3
G16	UI 7	39	0.4	-0.2	0.02	mA	1-3
G17	MP TO MPC	43	0.4	-0.6	0.02	mA	1-3
G18	MAINT ENBL	48	0.4	-0.2	0.02	mA	1-3
G19	MICRO CONT	49	0.4	-0.6	0.02	mA	1-3
G20	REQ IOCA 7	52	0.4	-1.26	-0.54	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	REQ IOCA 6	53	0.4	-1.26	-0.54	mA	1-3
G22	REQ IOCA 5	54					
G23	REQ IOCA 4	55					
G24	REQ IOCA 3	56					
G25	REQ IOCA 2	57					
G26	REQ IOCA 1	58					
G27	REQ IOCA 0	59					
G28	REQ IOCA 15	61					
G29	REQ IOCA 14	62					
G30	REQ IOCA 13	63					
G31	REQ IOCA 12	64					
G32	REQ IOCA 11	65					
G33	REQ IOCA 10	66					
G34	REQ IOCA 9	67					
G35	REQ IOCA 8	68	0.4	-1.26	-0.54	mA	1-3
G36	BUS 7	87	0.4	-0.25	0.07	mA	1-3
G37	BUS 6	88					
G38	BUS 5	89					
G39	BUS 4	91	0.4	-0.25	0.07	mA	1-3
G40	SOURCE IOC	92	0.4	-1.2	0.04	mA	1-3
G41	BUS 3	93	0.4	-0.25	0.07	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G42	BUS 2	94	0.4	-0.25	0.07	mA	1-3
G43	BUS 1	95	"	"	"	"	"
G44	BUS 0	96	0.4	-0.25	0.07	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G44
Power supply current limit	1.05 A	NA	G1-G44
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G44
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G44
low, (VL)	0.0 V	± 100 mV	G1-G44
Input current measurement	NA	± 11 uA	G1
	NA	± 31 uA	G2,G3
	NA	± 102 uA	G4
	NA	± 510 uA	G5
	NA	± 31 uA	G6
	NA	± 62 uA	G7
	NA	± 31 uA	G8
	NA	± 480 uA	G9

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 62 uA	G10
	NA	± 32 uA	G11
	NA	± 42 uA	G12-G14
	NA	± 31 uA	G15
	NA	± 11 uA	G16
	NA	± 31 uA	G17
	NA	± 11 uA	G18
	NA	± 31 uA	G19
	NA	± 36 uA	G20-G35
	NA	± 16 uA	G36-G39
	NA	± 62 uA	G40
	NA	± 16 uA	G41-G44

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	CHAN/INT	2	-0.4	2.4	4.5	V	1-4
H2	DATA SCAN	3					
H3	T/R	4					
H4	T1	5					
H5	T2	6					
H6	T3	7					
H7	T4	8					
H8	ENBL T1	12					
H9	CLOCK RATE	16	-0.4	2.4	4.5	V	1-4
H10	SEL T4	17	-0.16	2.4	4.5	V	1-4
H11	ENBL 250NS	22	-0.4	2.4	4.5	V	1-4
H12	MPC MODE	23					
H13	INIT SCAN	24	-0.4	2.4	4.5	V	1-4
H14	COMMON BUS	27	-0.16	2.4	4.5	V	1-4
H15	ENABLE RMW	28					
H16	ENBL BITE	29					
H17	IOC LPBACK	30	-0.16	2.4	4.5	V	1-4
H18	HIT (CM)	32	-0.4	2.4	4.5	V	1-4
H19	MPC MCLR	33	-0.16	2.4	4.5	V	1-4
H20	CHAIN/INT	41					
H21	DATA HIT	42	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H22	CHAN SEL 2	44	-0.16	2.4	4.5	V	1-4
H23	CHAN SEL 3	45					
H24	CHAN SEL 1	46					
H25	CHAN SEL 0	47	-0.16	2.4	4.5	V	1-4
H26	EN IOC MIC	50	-0.4	2.4	4.5	V	1-4
H27	EN IOCA 0	69					
H28	EN IOCA 1	70					
H29	EN IOCA 2	71					
H30	EN IOCA 3	72					
H31	EN IOCA 4	73					
H32	EN IOCA 5	74					
H33	EN IOCA 6	76					
H34	EN IOCA 7	77					
H35	EN IOCA 14	78					
H36	EN IOCA 15	79					
H37	EN IOCA 13	80					
H38	EN IOCA 12	82					
H39	EN IOCA 11	83					
H40	EN IOCA 10	84					
H41	EN IOCA 9	85					
H42	EN IOCA 8	86	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H43	BUS 7	87	-0.4	2.4	4.5	V	1-4
H44	BUS 6	88					
H45	BUS 5	89					
H46	BUS 4	91					
H47	BUS 3	93					
H48	BUS 2	94					
H49	BUS 1	95					
H50	BUS 0	96	-0.4	2.4	4.5	V	1-4
H51	CHAN SEL 0	97	-0.16	2.4	4.5	V	1-4
H52	CHAN SEL 1	98					
H53	CHAN SEL 2	99					
H54	CHAN SEL 3	100	-0.16	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H54
Power supply current limit	1.05 A	NA	H1-H54
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H9
	-0.16 mA	± 16 uA	H10
	-0.4 mA	± 40 uA	H11-H13
	-0.16 mA	± 16 uA	H14-H17
	-0.4 mA	± 40 uA	H18
	-0.16 mA	± 16 uA	H19-H25
	-0.4 mA	± 40 uA	H26-H50
	-0.16 mA	± 16 uA	H51-H54
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H54
low, (VL)	0.0 V	± 100 mV	H1-H54
Output voltage measurement	NA	± 105 mV	H1-H54

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

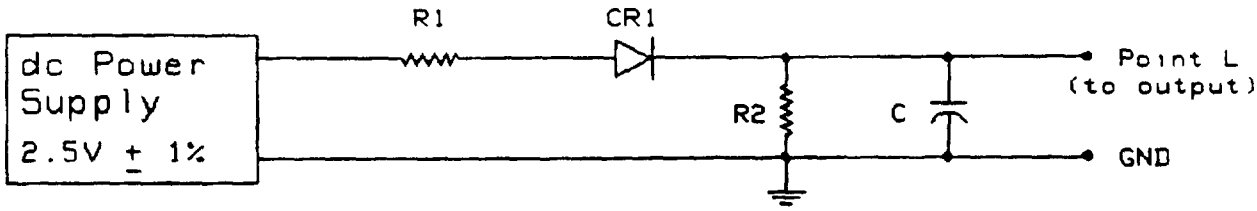
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,5,6,7,8,12, 16,22,23,24,32, 50,69,70,71,72, 73,74,76,77,78, 79,80,82,83,84, 85,86,87,88,89, 91,93,94,95,96	6200	6.0	0.5
17,27,28,29,30, 33,41,42,44,45, 46,47,97,98,99, 100	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	CHAN/INT	2	6.0	0.0	0.5	V	1-4
J2	DATA SCAN	3					
J3	T/R	4					
J4	T1	5					
J5	T2	6					
J6	T3	7					
J7	T4	8					
J8	ENBL T1	12					
J9	CLOCK RATE	16	6.0	0.0	0.5	V	1-4
J10	SEL T4	17	1.6	0.0	0.5	V	1-4
J11	ENBL 250NS	22	6.0	0.0	0.5	V	1-4
J12	MPC MODE	23					
J13	INIT SCAN	24	6.0	0.0	0.5	V	1-4
J14	COMMON BUS	27	1.6	0.0	0.5	V	1-4
J15	ENABLE RMW	28					
J16	ENBL BITE	29					
J17	IOC LPBACK	30	1.6	0.0	0.5	V	1-4
J18	HIT (CM)	32	6.0	0.0	0.5	V	1-4
J19	MPC MCLR	33	1.6	0.0	0.5	V	1-4
J20	CHAIN/INT	41					
J21	DATA HIT	42	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	CHAN SEL 2	44	1.6	0.0	0.5	V	1-4
J23	CHAN SEL 3	45					
J24	CHAN SEL 1	46					
J25	CHAN SEL 0	47	1.6	0.0	0.5	V	1-4
J26	EN IOC MIC	50	6.0	0.0	0.5	V	1-4
J27	EN IOCA 0	69					
J28	EN IOCA 1	70					
J29	EN IOCA 2	71					
J30	EN IOCA 3	72					
J31	EN IOCA 4	73					
J32	EN IOCA 5	74					
J33	EN IOCA 6	76					
J34	EN IOCA 7	77					
J35	EN IOCA 14	78					
J36	EN IOCA 15	79					
J37	EN IOCA 13	80					
J38	EN IOCA 12	82					
J39	EN IOCA 11	83					
J40	EN IOCA 10	84					
J41	EN IOCA 9	85					
J42	EN IOCA 8	86	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J43	BUS 7	87	6.0	0.0	0.5	V	1-4
J44	BUS 6	88					
J45	BUS 5	89					
J46	BUS 4	91					
J47	BUS 3	93					
J48	BUS 2	94					
J49	BUS 1	95					
J50	BUS 0	96	6.0	0.0	0.5	V	1-4
J51	CHAN SEL 0	97	1.6	0.0	0.5	V	1-4
J52	CHAN SEL 1	98					
J53	CHAN SEL 2	99					
J54	CHAN SEL 3	100	1.6	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:23Z
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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J54
Power supply current limit	1.05 A	NA	J1-J54
Forcing current, (Iol)	6.0 mA	± 600 uA	J1-J9
	1.6 mA	± 160 uA	J10
	6.0 mA	± 600 uA	J11-J13
	1.6 mA	± 160 uA	J14-J17
	6.0 mA	± 600 uA	J18
	1.6 mA	± 160 uA	J19-J25
	6.0 mA	± 600 uA	J26-J50
	1.6 mA	± 160 uA	J51-J54
Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J54
low, (VL)	0.0 V	± 100 mV	J1-J54
Output voltage measurement	NA	± 25 mV	J1-J54

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	COMM PEND	9	-18.0	-1.5	0.0	V	1-3
L2	CLASS I+C	11	-18.0	-1.2	0.0	V	1-3
L3	DUAL FIRST	13					
L4	BUS BIT 7	14					
L5	IO CLOCK	15					
L6	MICRO CIII	18					
L7	CHAN REQ	19					
L8	DEST CHAN	20					
L9	MACLR	21					
L10	UI 8	26	-18.0	-1.2	0.0	V	1-3
L11	UI 0	34	-18.0	-1.5	0.0	V	1-3
L12	UI 4	35					
L13	UI 5	36					
L14	UI 6	37	-18.0	-1.5	0.0	V	1-3
L15	SCE A 14	38	-18.0	-1.2	0.0	V	1-3
L16	UI 7	39	-18.0	-1.5	0.0	V	1-3
L17	MP TO MPC	43	-18.0	-1.2	0.0	V	1-3
L18	MAINT ENBL	48	-18.0	-1.5	0.0	V	1-3
L19	MICRO CONT	49	-18.0	-1.2	0.0	V	1-3
L20	SOURCE IOC	92	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L20
Power supply current limit	NA	NA	L1-L20
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L20
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L20 L1-L20
Input voltage measurement	NA NA NA NA NA NA NA NA	± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	L1 L2-L10 L11-L14 L15 L16 L17 L18 L19,L20

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	8.4	15.6	uF	1-4
M2	PSFC	51	7.0	13.0	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 360 nF ± 300 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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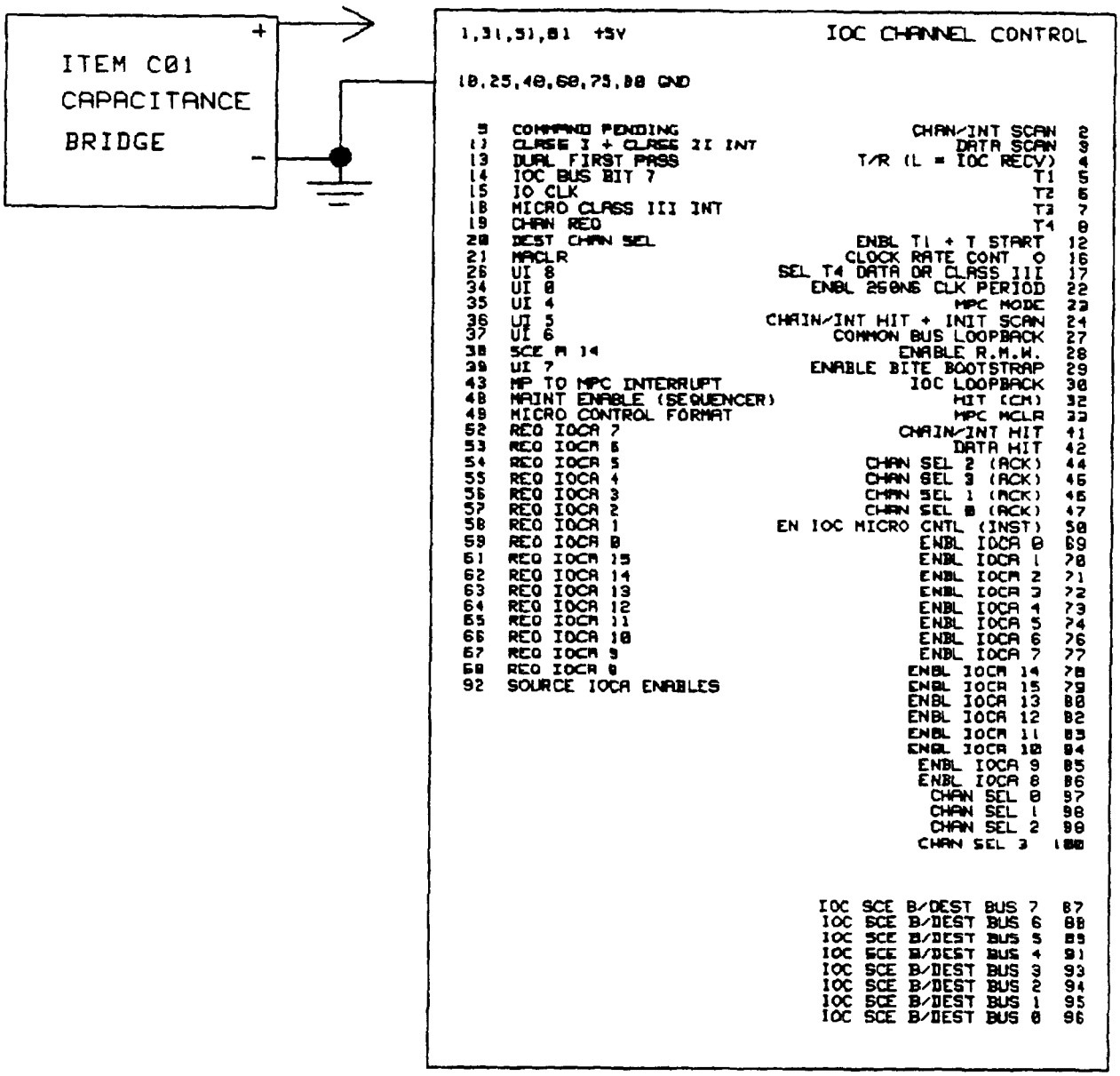


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	290	700	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.05 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 20.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.05 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: 2.4 V $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ 0.5 V

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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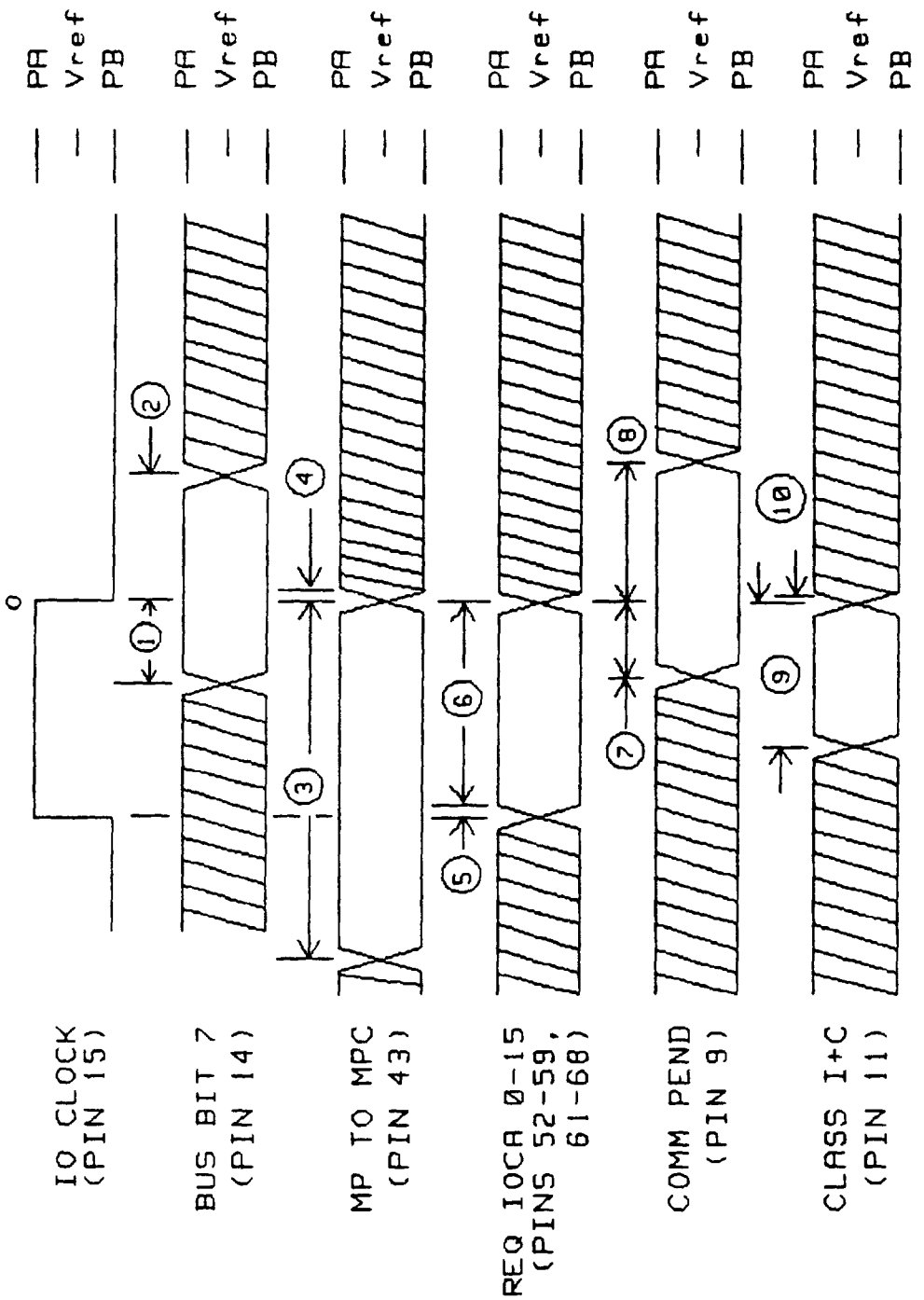


FIGURE 11. Input timing requirements for table XIV.

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NOTES:

1. (1) BUS BIT 7 to IO CLOCK. Setup time: minimum 23 nS.
2. (2) IO CLOCK to BUS BIT 7. Hold time: minimum 11 nS.
3. (3) MP TO MPC to IO CLOCK. Setup time: minimum 88 nS.
4. (4) IO CLOCK to MP TO MPC. Hold time: minimum 4 nS.
5. (5) REQ IOCA 0-15 to IO CLOCK. Setup time: minimum 6 nS.
6. (6) IO CLOCK to REQ IOCA 0-15. Hold time: minimum 43 nS.
7. (7) COMM PEND to IO CLOCK. Setup time: minimum 12 nS.
8. (8) IO CLOCK to COMM PEND. Hold time: minimum 14 nS.
9. (9) CLASS I+C to IO CLOCK. Setup time: minimum 23 nS.
10. (10) IO CLOCK to CLASS I+C. Hold time: minimum 6 nS.

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TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.05 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

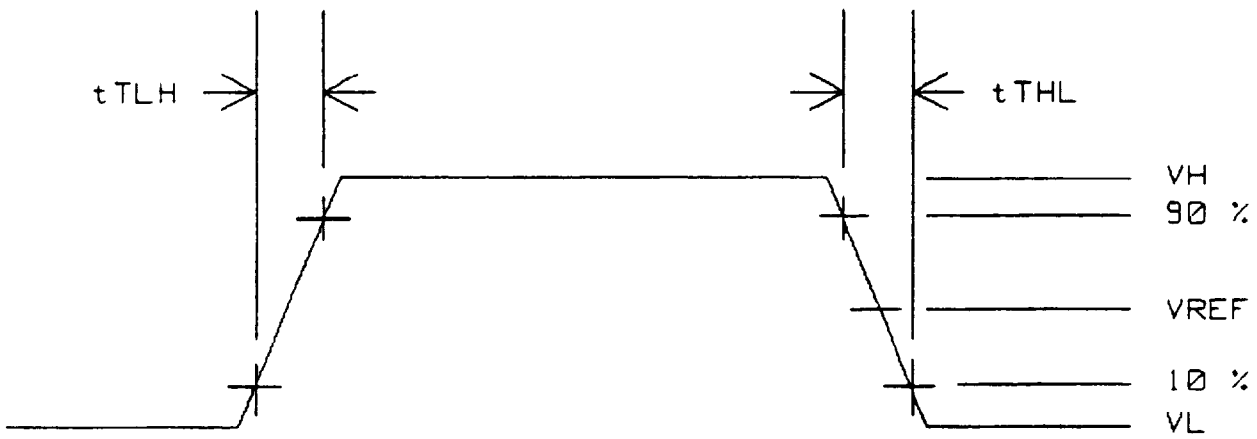
(b) The output compare levels shall be as follows:

High level: Vref ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ Vref

(c) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as specified on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	BUS BIT 7 to IO CLOCK	23	23	23	-	-	-	nS	12
2	th	IO CLOCK to BUS BIT 7	11	11	11	-	-	-	nS	12
3	tsu	MP TO MPC to IO CLOCK	88	88	88	-	-	-	nS	12
4	th	IO CLOCK to MP TO MPC	4	4	4	-	-	-	nS	12
5	tsu	REQ IOCA 0 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 1 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 2 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 3 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 4 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 5 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 6 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 7 to IO CLOCK	6	6	6	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	REQ IOCA 8 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 9 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 10 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 11 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 12 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 13 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 14 to IO CLOCK	6	6	6	-	-	-	nS	12
5	tsu	REQ IOCA 15 to IO CLOCK	6	6	6	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 0	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 1	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 2	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 3	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 4	43	43	43	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	th	IO CLOCK to REQ IOCA 5	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 6	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 7	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 8	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 9	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 10	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 11	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 12	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 13	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 14	43	43	43	-	-	-	nS	12
6	th	IO CLOCK to REQ IOCA 15	43	43	43	-	-	-	nS	12
7	tsu	COMM PEND to IO CLOCK	12	12	12	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	th	IO CLOCK to COMM PEND	14	14	14	-	-	-	nS	12
9	tsu	CLASS I+C to IO CLOCK	23	23	23	-	-	-	nS	12
10	th	IO CLOCK to CLASS I+C	6	6	6	-	-	-	nS	12
11	tpd	IO CLOCK to T1	-	-	-	64	64	64	nS	12
12	tpd	IO CLOCK to EN IOCA 0	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 1	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 2	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 3	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 4	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 5	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 6	-	-	-	96	96	96	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	IO CLOCK to EN IOCA 7	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 8	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 9	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 10	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 11	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 12	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 13	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 14	-	-	-	96	96	96	nS	12
12	tpd	IO CLOCK to EN IOCA 15	-	-	-	96	96	96	nS	12
13	tpd	IO CLOCK to T/R	-	-	-	90	90	90	nS	12
14	tpd	IO CLOCK to MPC MODE	-	-	-	64	64	64	nS	12
15	tpd	UI 0 to SEL T4	-	-	-	58	58	58	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
15	tpd	UI 4 to SEL T4	-	-	-	58	58	58	nS	12
15	tpd	UI 5 to SEL T4	-	-	-	58	58	58	nS	12
15	tpd	UI 6 to SEL T4	-	-	-	58	58	58	nS	12
16	tpd	IO CLOCK to DATA SCAN	-	-	-	79	79	79	nS	12
16	tpd	IO CLOCK to CHAN/INT	-	-	-	79	79	79	nS	12
17	tpd	SCE A 14 to EN IOC MIC	-	-	-	37	37	37	nS	12

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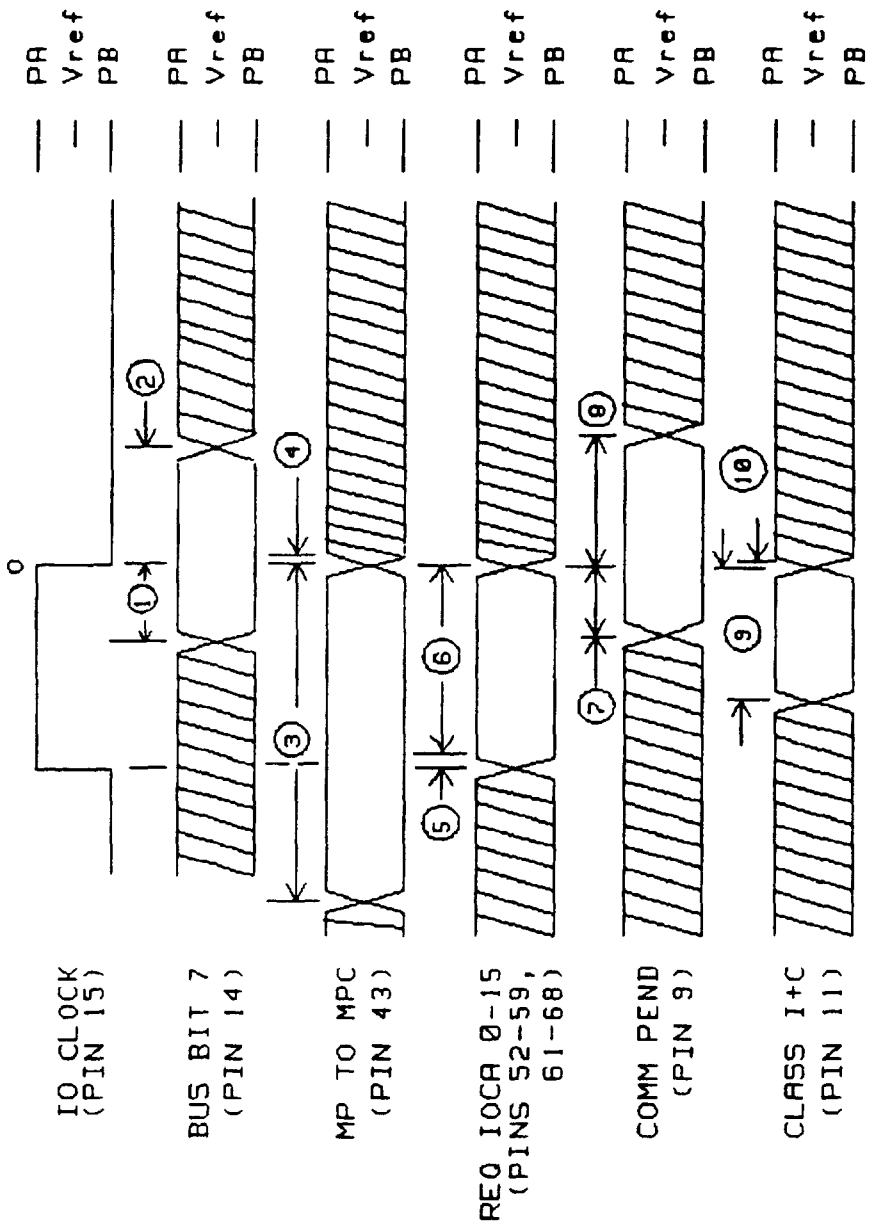


FIGURE 12a. Dynamic waveforms for table XVI.

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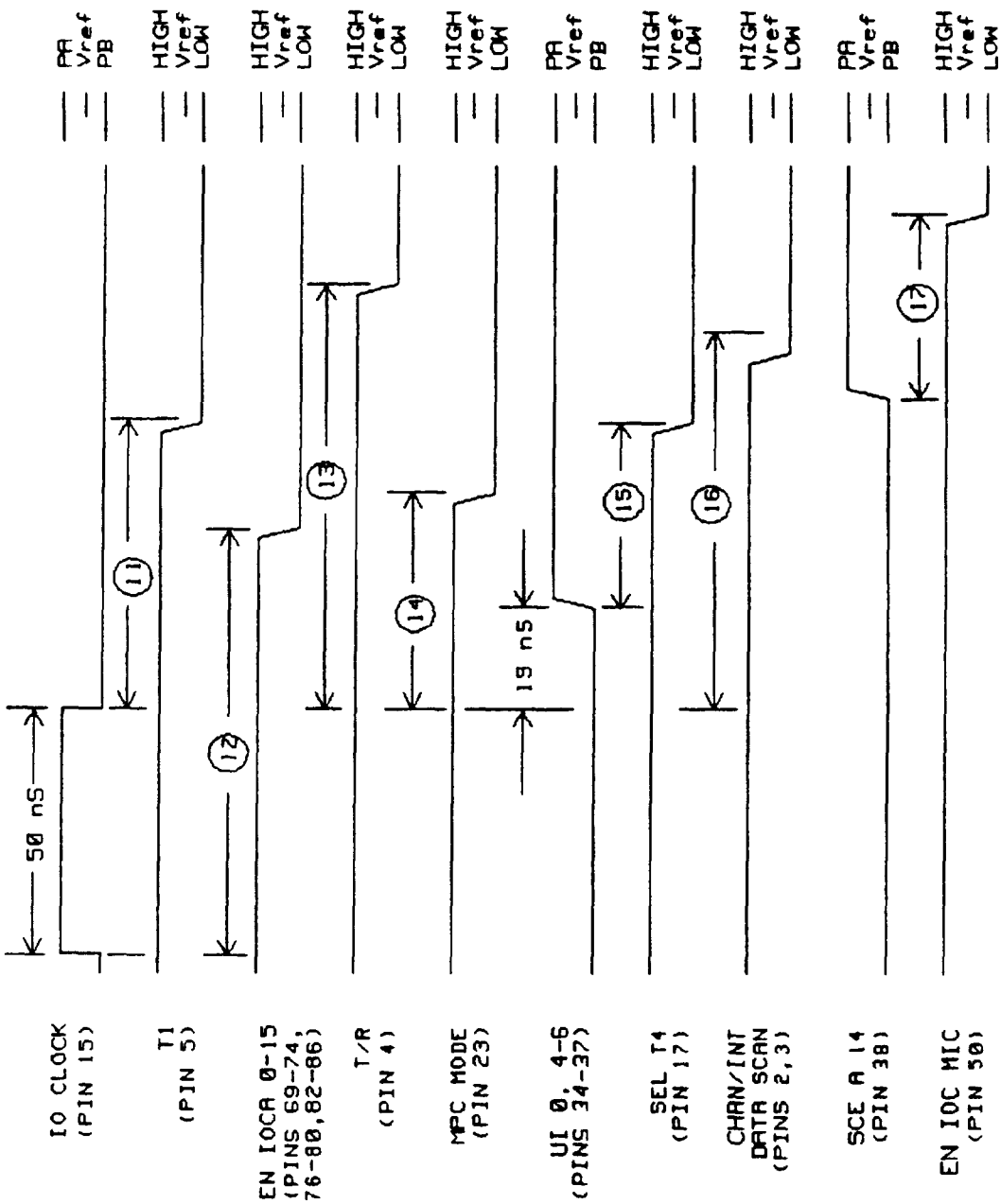


FIGURE 12b. Dynamic waveforms for table XVI.

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TABLE XVII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	COMM PEND	9	2	75	pF	1-4
R2	CLASS I+C	11	2	125	pF	1-4
R3	DUAL FIRST	13	2	75	pF	1-4
R4	BUS BIT 7	14	2	50	pF	1-4
R5	IO CLOCK	15	10	100	pF	1-4
R6	MICRO CIII	18	2	75	pF	1-4
R7	CHAN REQ	19	4	75	pF	1-4
R8	DEST CHAN	20	2	75	pF	1-4
R9	MACLR	21	12	150	pF	1-4
R10	UI 8	26	2	75	pF	1-4
R11	UI 0	34	4	75	pF	1-4
R12	UI 4	35				
R13	UI 5	36				
R14	UI 6	37	4	75	pF	1-4
R15	SCE A 14	38	2	75	pF	1-4
R16	UI 7	39	2	50	pF	1-4
R17	MP TO MPC	43	2	75	pF	1-4
R18	MAINT ENBL	48	2	50	pF	1-4
R19	MICRO CONT	49	2	75	pF	1-4
R20	REQ IOCA 7	52	4	50	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	REQ IOCA 6	53	4	75	pF	1-4
R22	REQ IOCA 5	54	4	50	pF	1-4
R23	REQ IOCA 4	55	4	75	pF	1-4
R24	REQ IOCA 3	56				
R25	REQ IOCA 2	57	4	75	pF	1-4
R26	REQ IOCA 1	58	4	50	pF	1-4
R27	REQ IOCA 0	59				
R28	REQ IOCA 15	61				
R29	REQ IOCA 14	62				
R30	REQ IOCA 13	63				
R31	REQ IOCA 12	64				
R32	REQ IOCA 11	65	4	50	pF	1-4
R33	REQ IOCA 10	66	4	75	pF	1-4
R34	REQ IOCA 9	67				
R35	REQ IOCA 8	68				
R36	SOURCE IOC	92	4	75	pF	1-4
Bidirectionals						
R37	BUS 7	87	4	75	pF	1-4
R38	BUS 6	88				
R39	BUS 5	89	4	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R40	BUS 4	91	4	75	pF	1-4
R41	BUS 3	93				
R42	BUS 2	94				
R43	BUS 1	95				
R44	BUS 0	96	4	75	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R44
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R44
Capacitance measurement	NA	± 3.65 pF	R1
	NA	± 6.15 pF	R2
	NA	± 3.65 pF	R3
	NA	± 2.4 pF	R4
	NA	± 4.5 pF	R5
	NA	± 3.65 pF	R6
	NA	± 3.55 pF	R7
	NA	± 3.65 pF	R8
	NA	± 6.9 pF	R9
	NA	± 3.65 pF	R10
	NA	± 3.55 pF	R11-R14

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 3.65 pF	R15
	NA	± 2.4 pF	R16
	NA	± 3.65 pF	R17
	NA	± 2.4 pF	R18
	NA	± 3.65 pF	R19
	NA	± 2.3 pF	R20
	NA	± 3.55 pF	R21
	NA	± 2.3 pF	R22
	NA	± 3.55 pF	R23-R254
	NA	± 2.3 pF	R26-R32
	NA	± 3.55 pF	R33-R44

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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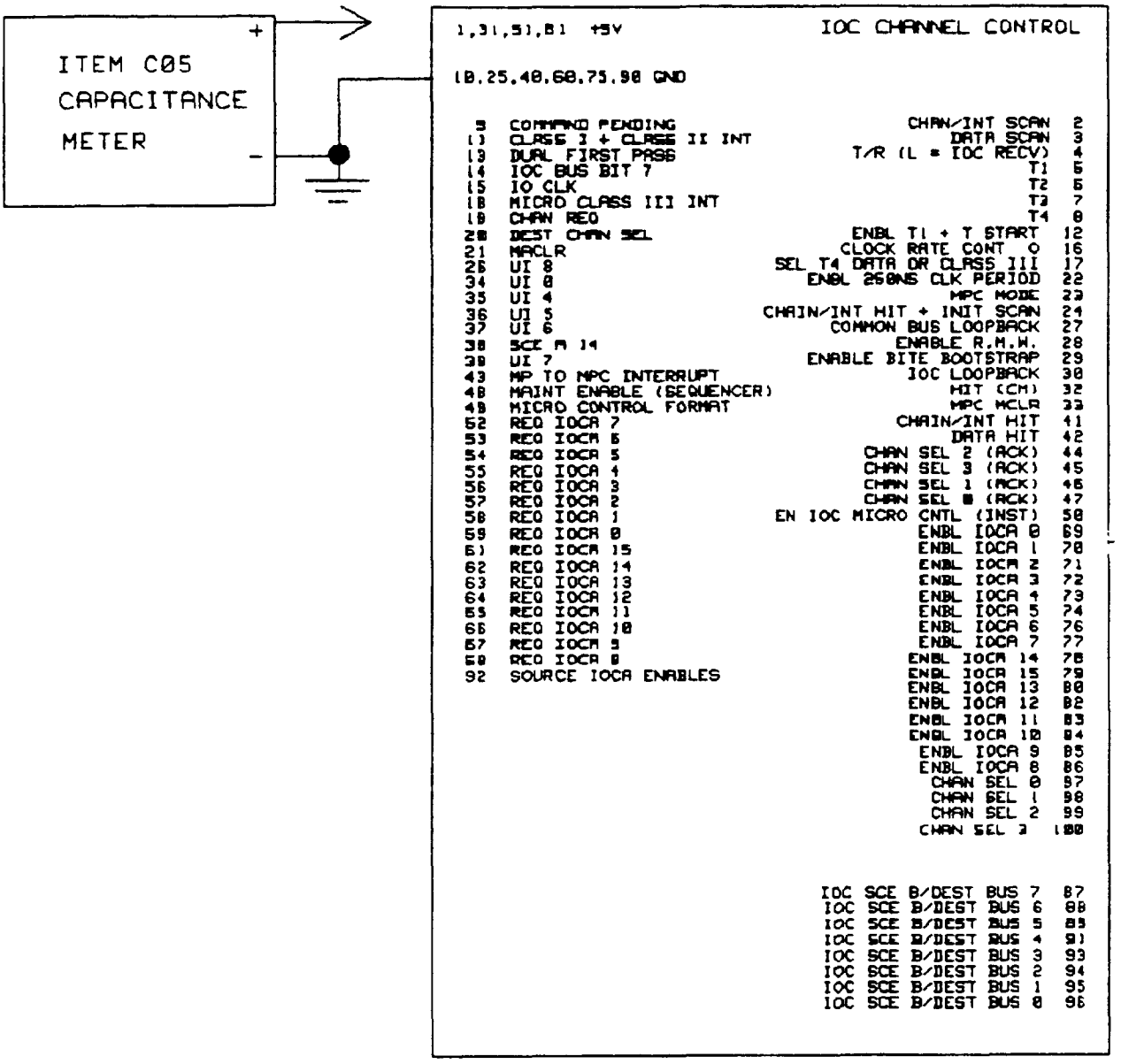


FIGURE 13. Input and output capacitance test circuit for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin number	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.05 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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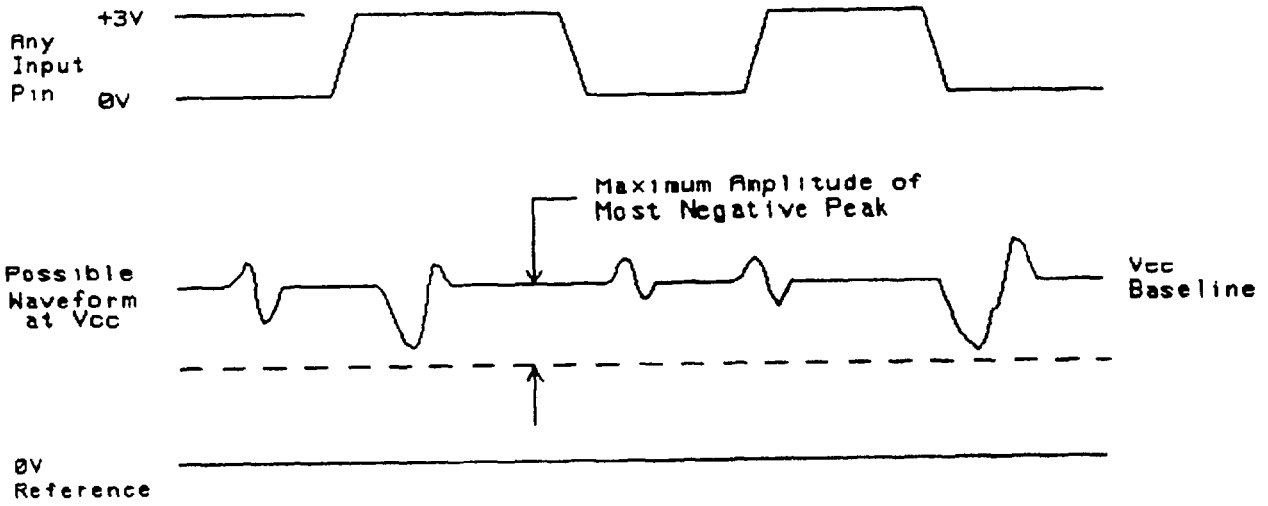
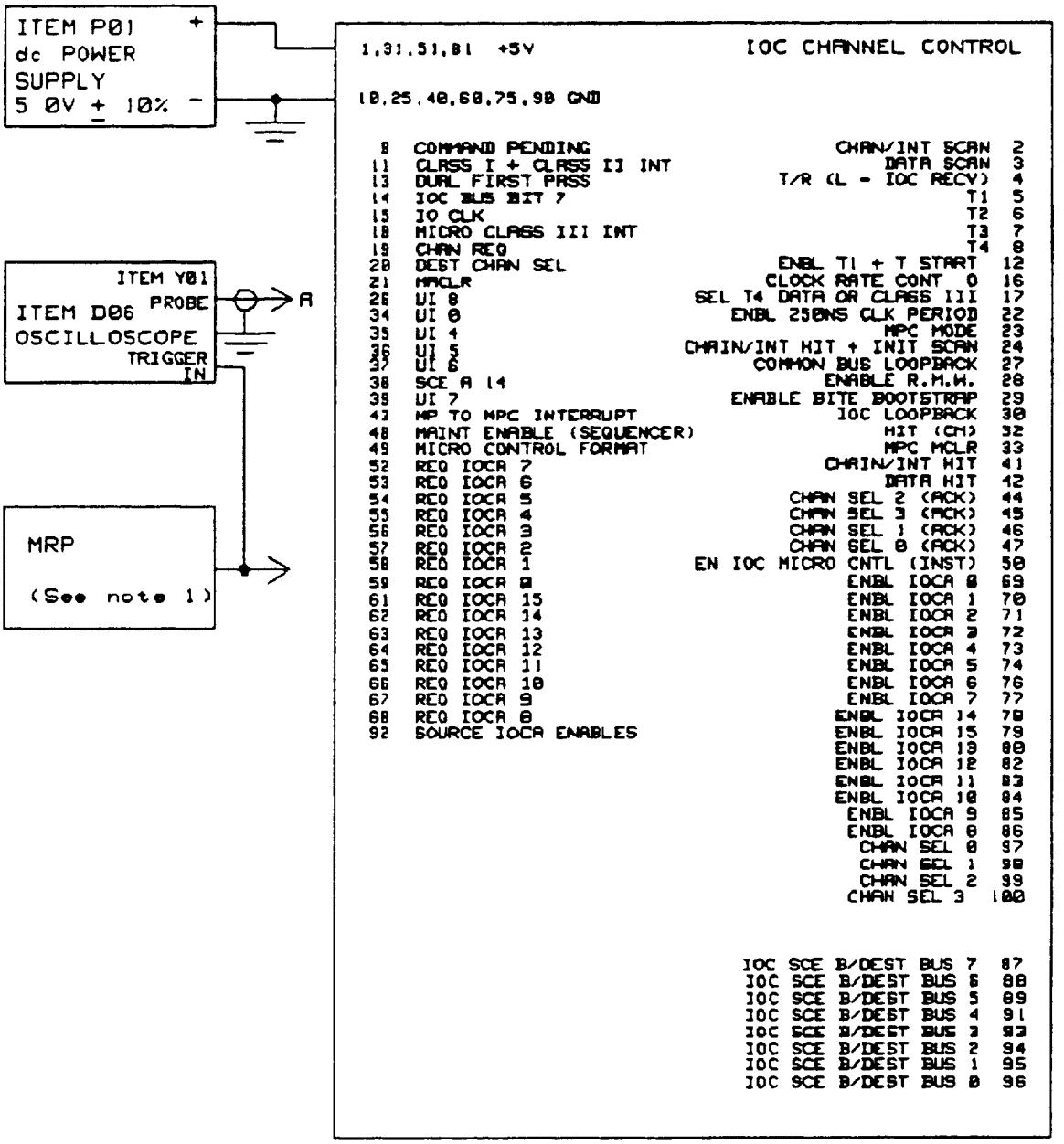


FIGURE 14. Power supply transient amplitude waveform for table XVIII.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XVIII.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-M-28787/506	Modules, Standard Electronic IOC Channel Control Module, Key Code AWH

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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